



AOS-CX 10.18.xxxx Maintenance mode Guide

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AOS-CX 10.18.xxxx Maintenance mode Guide (Internal use only)

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Aruba Maintenance Mode

This section provides comprehensive documentation for the Aruba Maintenance Mode feature CLI commands and REST API. Aruba Maintenance Mode is a feature designed to enable network operators to perform maintenance tasks, such as hardware replacements, EOS upgrades, re-cabling, or configuration changes, while minimizing disruption to live network traffic. When activated, maintenance mode uses protocols like OSPF and BGP to gracefully reroute traffic away from the switch or its components, ensuring minimal traffic impact and service continuity.

We will focus on CLI commands and operations, however, commands include corresponding REST API examples. For REST API developers, a comprehensive REST API terminology mapping and data model reference section is available in the NAME OF SECTION.

Design Principles and Traffic Impact

This section describes the underlying mechanisms, network requirements, and expected traffic behavior during maintenance mode operations. Understanding these principles is essential for proper deployment planning and setting appropriate expectations for traffic impact.

Underlying Protocol Mechanisms

Maintenance mode leverages standard protocol mechanisms to signal maintenance state to the network and gracefully redirect traffic away from the switch:

	OSPF Max-Metric	BGP Graceful Shutdown (GSHUT)
Mechanism	Maximum metric values (0xFFFF) are applied to OSPF interface costs in Link State Advertisements (LSAs).	BGP Graceful Shutdown (GSHUT) community (GRACEFUL_SHUTDOWN 65535:0) is advertised with BGP routes, followed by BGP neighbor session shutdown after a configurable shutdown-delay-timer.
Behavior	• OSPF neighbors receive LSAs advertising maximum metric paths through the switch • Neighboring routers recalculate SPF (Shortest Path First) with the new metric values • Traffic is redirected through alternate lower-cost paths if available	• GSHUT community (65535:0) is added to all advertised BGP routes • IBGP peers receive routes with GSHUT community and lower route preference • IBGP Peers select alternate paths if available, avoiding routes through the maintenance switch • EBGP peer receive routes with only GSHUT community

	OSPF Max-Metric	BGP Graceful Shutdown (GSHUT)
	OSPF neighbor relationships remain active; only metric values change	- EBGP peers should be configured with route-map actions as, when GSHUT community is received route preference need to be altered - After GSHUT community advertisement, the configurable shutdown-delay-timer (default 120 seconds) begins - BGP neighbor sessions are brought down gracefully after the shutdown-delay-timer expires - Session teardown allows peers to immediately remove routes and complete convergence
Timing	OSPF reconvergence depends on LSA flooding, SPF calculation, and RIB/FIB updates across the network.	Two-phase process: - GSHUT community advertisement triggers peer-side route preference changes - Shutdown-delay-timer expiration (configurable, default 120 seconds) triggers session teardown and final route removal

Critical Network Topology Requirements

Redundant Path Requirement

Maintenance mode requires redundant network paths for traffic redirection. The network topology **must** provide alternate paths for traffic to avoid the maintenance switch.



NOTE

Why redundancy is essential?

- Maintenance mode signals protocols to deprioritize paths through the switch
- Without alternate paths, traffic has nowhere to redirect
- Protocol mechanisms (max-metric, GSHUT) make the switch less preferred, not unreachable

Topology Planning:

1. **OSPF Networks:** Ensure mesh or redundant ring topologies with multiple paths between endpoints
2. **BGP Networks:** Verify multiple BGP paths exist for critical prefixes before maintenance
3. **Validation:** Use network simulation or pre-maintenance path analysis to confirm redundancy

Single Point of Failure Warning: If the maintenance switch is a single point of failure (no alternate path), maintenance mode **cannot** prevent traffic disruption. Traffic loss will occur regardless of protocol signaling. While OSPF routes will advertise max-metric and BGP neighbor sessions will be shut down, the switch itself remains reachable for management and control plane operations. Maintenance mode affects only the routing protocols, not the underlying switch management accessibility.

Traffic Impact Expectations

Activation impact on traffic

When maintenance mode is activated, the switch signals protocols to redirect traffic. Traffic loss is minimized but not eliminated.

Traffic loss considerations:

- **Not Guaranteed Zero Loss:** Brief traffic disruption may occur during protocol reconvergence
- **Convergence Windows:** OSPF SPF recalculation and BGP path selection take time
- **Micro-Outages:** Sub-second to few-second interruptions possible during switchover
- **Flow Disruption:** Existing flows may experience packet loss during path transition
- **New Flows:** New flows are immediately redirected once protocols converge

Factors affecting traffic Loss:

1. **Protocol Convergence Speed:** Faster OSPF timers and BGP PIC (Prefix Independent Convergence) reduce loss
2. **Network Topology:** More alternate paths and better-meshed networks converge faster
3. **Traffic Patterns:** Elephant flows and stateful connections experience more disruption
4. **Hardware Capabilities:** Switch forwarding plane performance during reconvergence

Best-Effort Traffic Minimization: Maintenance mode uses industry-standard graceful mechanisms, but cannot guarantee hitless operation. Plan maintenance windows during low-traffic periods.

Deactivation impact on traffic



NOTE

Deactivating maintenance mode with the `no maintenance-mode activate` command **can cause traffic loss** during BGP neighbor reconvergence.

Why does deactivation cause traffic loss?:

- **BGP Session Restoration:** BGP neighbors were shut down during maintenance mode
- **Session Re-establishment:** TCP sessions must reconnect, BGP FSM transitions through states
- **Route Re-advertisement:** All routes must be re-exchanged and processed
- **RIB/FIB Updates:** Routing tables and forwarding tables are updated with new paths
- **Reconvergence Window:** Significant time (seconds to minutes) for full BGP reconvergence

Traffic Disruption During Deactivation:

- Traffic that would use BGP routes through the switch experiences loss until BGP reconverges
- Existing flows through alternate paths may shift back, causing brief interruption
- OSPF reconvergence (max-metric removal) is typically faster than BGP

Deactivation Best Practices:

1. **Plan Deactivation Windows:** Schedule deactivation during maintenance windows, not production hours
2. **Monitor Reconvergence:** Watch BGP session states, route counts, and traffic flows during deactivation
3. **Staged Deactivation:** For large networks, consider deactivating in stages if supported by topology
4. **Avoid Rapid Cycling:** Do not repeatedly activate/deactivate; each cycle causes reconvergence disruption

The following table summarizes the impact on traffic:

Operation	Traffic Impact	Primary Cause	Duration
Activation	Minimized, not eliminated	Protocol reconvergence to alternate paths	Sub-second to few seconds
Deactivation	Expected loss during BGP reconvergence	BGP session re-establishment and route re-advertisement	Seconds to minutes
Steady State (Active)	Minimal, traffic on alternate paths	Normal forwarding through alternate topology	N/A (stable)

Deployment Planning Checklist

Before activating maintenance mode in production networks:

- **Verify Redundant Paths:** Confirm alternate paths exist for all critical traffic flows
- **Test in Lab:** Validate maintenance mode behavior in lab topology matching production
- **Plan Maintenance Windows:** Schedule activation and deactivation during low-traffic periods
- **Monitor Protocols:** Prepare monitoring for OSPF metrics and BGP paths during operation
- **Document Expectations:** Set stakeholder expectations about potential micro-outages during activation
- **Plan Deactivation Carefully:** Treat deactivation as a potentially disruptive event requiring monitoring
- **Have Rollback Plan:** Prepare to quickly deactivate if unexpected traffic loss occurs

Configuration Examples

This section provides comprehensive examples showing how to configure and use maintenance mode with protocol-specific maintenance-units.

Protocol-Specific maintenance-units Architecture

Maintenance mode uses protocol-specific maintenance-units where each maintenance-unit is dedicated to a single protocol type:

Protocol	Description
BGP maintenance-units	Configure BGP graceful shutdown behavior.
OSPFv2/OSPFv3/OSPFv3-AF maintenance-units	Configure OSPF max-metric behavior for specific OS PF variants.

Key Design Principle: Maintenance-units are protocol-specific and cannot contain multiple protocols. To affect multiple protocols during maintenance, create separate protocol-specific maintenance-units and apply all of them to a maintenance mode profile.

Organizational Flexibility: You can organize maintenance-units by function, such as, "campus-bgp", "campus-ospfv2" or by network layer, for example, "underlay-ospfv2", "overlay-bgp". Both approaches run identically; the choice is purely organizational for configuration clarity.

Subtopics

Alternative Profile Configurations

Example 1: Complete Maintenance Configuration

Example 2: Layer-Separated Feature-Sets Configuration

Alternative Profile Configurations

This example demonstrates how to use protocol-specific maintenance-units organized by network layer for maintenance operations that require coordinated traffic redirection across underlay and overlay protocols.

Since only one profile can exist at a time, you can reconfigure for different maintenance scenarios:

For Underlay-Only Maintenance

```
switch(config) # no maintenance-mode profile complete-profile
switch(config) # maintenance-mode profile underlay-profile
switch(config-maint-mode-underlay-profile) # maintenance-unit apply underlay-ospfv2
switch(config-maint-mode-underlay-profile) # maintenance-unit apply underlay-ospfv3
switch(config-maint-mode-underlay-profile) # exit
switch(config) # maintenance-mode activate profile underlay-profile
```

For Overlay-Only Maintenance

```
switch(config) # no maintenance-mode profile underlay-profile
switch(config) # maintenance-mode profile overlay-profile
switch(config-maint-mode-overlay-profile) # maintenance-unit apply overlay-bgp
switch(config-maint-mode-overlay-profile) # exit
switch(config) # maintenance-mode activate profile overlay-profile
```

Example 1: Complete Maintenance Configuration

Configuration Overview

This example demonstrates configuring maintenance mode with protocol-specific maintenance-unit. We create separate BGP, OSPFv2, and OSPFv3 maintenance-units, then apply all of them to a single maintenance mode profile for coordinated maintenance affecting all network layers.

Complete Configuration Example

Create BGP maintenance-unit for overlay services:

```
switch(config) # maintenance-unit bgp campus-bgp
switch(config-maintenance-unit-bgp-campus-bgp) # shutdown-delay-timer 120
switch(config-maintenance-unit-bgp-campus-bgp) # neighbor 10.1.1.1 vrf red
switch(config-maintenance-unit-bgp-campus-bgp) # neighbor 10.1.1.2 vrf red
switch(config-maintenance-unit-bgp-campus-bgp) # exit
```

Create OSPFv2 maintenance-unit for IPv4 underlay routing:

```
switch(config)# maintenance-unit ospfv2 campus-ospfv2  
switch(config-maintenance-unit-ospfv2-campus-ospfv2)# process 1 vrf red  
switch(config-maintenance-unit-ospfv2-campus-ospfv2)# process 2 vrf red  
switch(config-maintenance-unit-ospfv2-campus-ospfv2)# exit
```

Create OSPFv3 maintenance-unit for IPv6 underlay routing:

```
switch(config)# maintenance-unit ospfv3 campus-ospfv3  
switch(config-maintenance-unit-ospfv3-campus-ospfv3)# process 1 vrf red  
switch(config-maintenance-unit-ospfv3-campus-ospfv3)# exit
```

Create maintenance mode profile and apply all maintenance-units:

```
switch(config)# maintenance-mode profile campus-profile  
switch(config-maint-mode-campus-profile)# maintenance-unit apply campus-bgp  
switch(config-maint-mode-campus-profile)# maintenance-unit apply campus-ospfv2  
switch(config-maint-mode-campus-profile)# maintenance-unit apply campus-ospfv3  
switch(config-maint-mode-campus-profile)# exit
```

Activate maintenance mode with the campus-profile:

```
switch(config)# maintenance-mode activate profile campus-profile
```

Show Running Configuration

```
switch# show running-config maintenance-mode  
# Feature-set definitions  
maintenance-unit bgp campus-bgp  
    shutdown-delay-timer 120  
    neighbor 10.1.1.1 vrf red  
    neighbor 10.1.1.2 vrf red  
  
maintenance-unit ospfv2 campus-ospfv2  
    process 1 vrf red  
    process 2 vrf red  
  
maintenance-unit ospfv3 campus-ospfv3  
    process 1 vrf red  
  
# Maintenance mode configuration  
maintenance-mode profile campus-profile  
    maintenance-unit apply campus-bgp  
    maintenance-unit apply campus-ospfv2
```

```
maintenance-unit apply campus-ospfv3
maintenance-mode activate profile campus-profile
```



NOTE

When viewing the complete switch configuration with **show running-config** command (without the maintenance-mode filter), the maintenance-mode section appears toward the end of the output, after most other configuration sections.

Execution Flow Explanation

Activation Flow

When maintenance mode is activated, all protocols from the applied maintenance-units are executed:

Feature-Set	Purpose	Action	Result
BGP (campus-bgp)	Gracefully remove the switch from BGP overlay services.	BGP neighbors 10.1.1.1 and 10.1.1.2 in VRF red are put into maintenance mode.	BGP graceful shutdown community is sent to peers and sessions are brought down to redirect overlay traffic.
OSPFv2 (campus-ospfv2)	Signal maintenance state for OSPFv2 IPv4 underlay routing.	OSPFv2 processes 1 and 2 in VRF red are put into maintenance mode.	OSPFv2 max-metric is applied to gracefully remove the switch from IPv4 underlay routing.
OSPFv3 (campus-ospfv3)	Signal maintenance state for OSPFv3 IPv6 underlay routing.	OSPFv3 process 1 in VRF red is put into maintenance mode.	OSPFv3 max-metric is applied to gracefully remove the switch from IPv6 underlay routing.

Deactivation Flow

When maintenance mode is deactivated using **no maintenance-mode activate**, normal operation is restored for all protocols from all applied maintenance-units. All maintenance mode configurations including the profile and maintenance-units remain configured and can be reactivated later.

REST API Configuration Example

This example demonstrates how to configure maintenance mode with protocol-specific maintenance-units that provide clear separation by protocol type, enabling granular control and reusability of configurations.

CLI protocol-specific maintenance-units map to two separate REST tables:

- maintenance-unit bgp → Feature_Set_BGP table
- maintenance-unit ospfv2/ospfv3/ospfv3-af → Feature_Set_OSPF table (one table for all OSPF variants)

Step 1: Create OSPFv2 maintenance-unit

```
curl -X POST "/rest/latest/system/feature_set_bgps" \
-H "Content-Type: application/json" \
-d '{
  "name": "campus-bgp",
  "bgp_peer_vrfs": {
    "10.1.1.1": "/rest/latest/system/vrfs/red",
    "10.1.1.2": "/rest/latest/system/vrfs/red"
  },
  "shutdown_delay_timer": 120,
  "description": "BGP overlay services"
}'
```

Step 2: Create OSPFv2 maintenance-unit:

```
curl -X POST "/rest/latest/system/feature_set_ospfs" \
-H "Content-Type: application/json" \
-d '{
  "name": "campus-ospfv2",
  "protocol_type": "ospfv2",
  "ospf_process_vrfs": {
    "1": "/rest/latest/system/vrfs/red",
    "2": "/rest/latest/system/vrfs/red"
  },
  "description": "OSPFv2 IPv4 underlay routing"
}'
```

Step 3: Create OSPFv3 maintenance-unit

```
curl -X POST "/rest/latest/system/feature_set_ospfs" \
-H "Content-Type: application/json" \
-d '{
  "name": "campus-ospfv3",
  "protocol_type": "ospfv3",
  "ospf_process_vrfs": {
    "1": "/rest/latest/system/vrfs/red"
  },
  "description": "OSPFv3 IPv6 underlay routing"
}'
```

Step 4: Create maintenance mode profile:

```
curl -X POST "/rest/latest/system/maintenance_mode_profiles" \
-H "Content-Type: application/json" \
-d '{
  "name": "campus-mode-profile",
  "description": "Campus network maintenance orchestration"
}'
```

Step 5: Create stage and link all maintenance-units to mode profile:

```
curl -X POST "/rest/latest/system/maintenance_mode_profiles/campus-mode-
profile/stages" \
-H "Content-Type: application/json" \
-d '{
  "stage_id": 1,
  "feature_set_bgp": ["/rest/latest/system/feature_set_bgps/campus-bgp"],
  "feature_set_ospf": [
    "/rest/latest/system/feature_set_ospfs/campus-ospfv2",
    "/rest/latest/system/feature_set_ospfs/campus-ospfv3"
  ]
}'
```

Step 6: Activate maintenance mode:

```
curl -X PATCH "/rest/latest/system/maintenance_mode_profiles/campus-mode-
profile" \
-H "Content-Type: application/json" \
-d '{"activated": true}'
```

Step 7: Monitor status:

```
curl -X GET "/rest/latest/system/maintenance_mode_profiles/campus-mode-
profile"
```

Step 8: Deactivate when maintenance complete:

```
curl -X PATCH "/rest/latest/system/maintenance_mode_profiles/campus-mode-
profile" \
-H "Content-Type: application/json" \
-d '{"activated": false}'
```

Example 2: Layer-Separated Feature-Sets Configuration

Configuration Overview

This example demonstrates organizing maintenance-units by network layer. We create separate "underlay" OSPFv2/OSPFv3 maintenance-units and an "overlay" BGP maintenance-unit, then apply them to a single profile for coordinated maintenance affecting all network layers.

Complete Configuration Example

Create underlay OSPFv2 maintenance-unit for IPv4 routing:

```
switch(config) # maintenance-unit ospfv2 underlay-ospfv2
switch(config-maintenance-unit-ospfv2-underlay-ospfv2) # process 1 vrf red
switch(config-maintenance-unit-ospfv2-underlay-ospfv2) # process 2 vrf red
switch(config-maintenance-unit-ospfv2-underlay-ospfv2) # exit
```

Create underlay OSPFv3 maintenance-unit for IPv6 routing:

```
switch(config)# maintenance-unit ospfv3 underlay-ospfv3  
switch(config-maintenance-unit-ospfv3-underlay-ospfv3)# process 1 vrf red  
switch(config-maintenance-unit-ospfv3-underlay-ospfv3)# exit
```

Create overlay BGP maintenance-unit for services:

```
switch(config)# maintenance-unit bgp overlay-bgp  
switch(config-maintenance-unit-bgp-overlay-bgp)# shutdown-delay-timer 120  
switch(config-maintenance-unit-bgp-overlay-bgp)# neighbor 10.1.1.1 vrf red  
switch(config-maintenance-unit-bgp-overlay-bgp)# neighbor 10.1.1.2 vrf red  
switch(config-maintenance-unit-bgp-overlay-bgp)# exit
```

Create profile applying all maintenance-units for complete maintenance

```
switch(config)# maintenance-mode profile complete-profile  
switch(config-maint-mode-complete-profile)# maintenance-unit apply underlay-ospfv2  
switch(config-maint-mode-complete-profile)# maintenance-unit apply underlay-ospfv3  
switch(config-maint-mode-complete-profile)# maintenance-unit apply overlay-bgp  
switch(config-maint-mode-complete-profile)# exit
```

Activate maintenance mode with the complete profile

```
switch(config)# maintenance-mode activate profile complete-profile
```

Show Running Configuration

```
switch# show running-config maintenance-mode  
# Feature-set definitions  
maintenance-unit ospfv2 underlay-ospfv2  
    process 1 vrf red  
    process 2 vrf red  
  
maintenance-unit ospfv3 underlay-ospfv3  
    process 1 vrf red  
  
maintenance-unit bgp overlay-bgp  
    shutdown-delay-timer 120  
    neighbor 10.1.1.1 vrf red  
    neighbor 10.1.1.2 vrf red  
  
# Maintenance mode profile configuration  
maintenance-mode profile complete-profile  
    maintenance-unit apply underlay-ospfv2  
    maintenance-unit apply underlay-ospfv3
```

```
maintenance-unit apply overlay-bgp
maintenance-mode activate profile complete-profile
```



NOTE

When viewing the complete switch configuration with **show running-config** (without the maintenance-mode filter), the maintenance-mode section appears toward the end of the output, after most other configuration sections.

Activation Flow Explanation

When maintenance mode is activated with **complete-profile**, all underlay and overlay maintenance-units are applied simultaneously:

Feature-Set	Purpose	Action	Result
Underlay OSPFv2 (underlay-ospfv2)	Signal maintenance state for OSPFv2 IPv4 underlay routing.	OSPFv2 processes 1 and 2 in VRF red are put into maintenance mode.	OSPFv2 max-metric is applied to gracefully redirect IPv4 underlay routing traffic away from the switch.
Underlay OSPFv3 (underlay-ospfv3)	Signal maintenance state for OSPFv3 IPv6 underlay routing.	OSPFv3 process 1 in VRF red is put into maintenance mode.	OSPFv3 max-metric is applied to gracefully redirect IPv6 underlay routing traffic away from the switch.

All configured routing protocols (underlay OSPF and overlay BGP) signal maintenance state simultaneously, providing coordinated traffic redirection across all network layers.

Deactivation Flow

When maintenance mode is deactivated using **no maintenance-mode activate**, normal operation is restored for all protocols in all maintenance-units. The profile and maintenance-units remain configured and can be reactivated later.

REST API Configuration Example

CLI protocol-specific maintenance-units map to two separate REST tables:

- maintenance-unit ospfv2/ospfv3 → Feature_Set_OSPF table (one table for all OSPF variants)
- maintenance-unit bgp → Feature_Set_BGP table

Step 1: Create underlay OSPFv2 maintenance-unit

```
curl -X POST "/rest/latest/system/feature_set_ospfs" \
-H "Content-Type: application/json" \
-d '{
```



```

    "name": "underlay-ospfv2",
    "protocol_type": "ospfv2",
    "ospf_process_vrfs": {
      "1": "/rest/latest/system/vrfs/red",
      "2": "/rest/latest/system/vrfs/red"
    },
    "description": "Underlay OSPFv2 IPv4 routing"
  }'

```

Step 2: Create underlay OSPFv3 maintenance-unit

```

curl -X POST "/rest/latest/system/feature_set_ospfs" \
-H "Content-Type: application/json" \
-d '{
  "name": "underlay-ospfv3",
  "protocol_type": "ospfv3",
  "ospf_process_vrfs": {
    "1": "/rest/latest/system/vrfs/red"
  },
  "description": "Underlay OSPFv3 IPv6 routing"
}'

```

Step 3: Create overlay BGP maintenance-unit

```

curl -X POST "/rest/latest/system/feature_set_bgps" \
-H "Content-Type: application/json" \
-d '{
  "name": "overlay-bgp",
  "bgp_peer_vrfs": {
    "10.1.1.1": "/rest/latest/system/vrfs/red",
    "10.1.1.2": "/rest/latest/system/vrfs/red"
  },
  "shutdown_delay_timer": 120,
  "description": "Overlay BGP services"
}'

```

Step 4: Create maintenance mode profile

```

curl -X POST "/rest/latest/system/maintenance_mode_profiles" \
-H "Content-Type: application/json" \
-d '{
  "name": "complete-mode-profile",
  "description": "Complete maintenance with underlay and overlay"
}'

```

Step 5: Create stage and link all maintenance-units to mode profile

```

curl -X POST "/rest/latest/system/maintenance_mode_profiles/complete-mode-profile/stages" \
-H "Content-Type: application/json" \

```

```
-d '{
  "stage_id": 1,
  "feature_set_bgp": ["/rest/latest/system/feature_set_bgps/overlay-bgp"],
  "feature_set_ospf": [
    "/rest/latest/system/feature_set_ospfs/underlay-ospfv2",
    "/rest/latest/system/feature_set_ospfs/underlay-ospfv3"
  ]
}
```

Step 6: Activate maintenance mode

```
Ocurl -X PATCH "/rest/latest/system/maintenance_mode_profiles/complete-mode-profile" \
-H "Content-Type: application/json" \
-d '{"activated": true}'
```

Step 7: Deactivate when complete

```
curl -X PATCH "/rest/latest/system/maintenance_mode_profiles/complete-mode-profile" \
-H "Content-Type: application/json" \
-d '{"activated": false}'
```

Configuration Notes

This layer-separated approach demonstrates:

1. **Layer Separation:** Protocol configurations organized by network layer (underlay OSPF vs. overlay BGP)
2. **Independent Protocol maintenance-units:** Each protocol has its own dedicated maintenance-unit
3. **Profile Flexibility:** Profile can be deleted and recreated with different maintenance-units combinations for different maintenance scenarios

Operational Behavior and Recommendations

This section provides important information about maintenance mode operational behavior, default configurations, feature interactions, and best practices for safe operation.

Default Profile Behavior

When no custom maintenance mode profile is configured, the system automatically applies a comprehensive default profile that affects all network protocols. This default profile uses protocol-specific default maintenance-units that are automatically created and applied, though they do not appear in the device configuration database.

Equivalent default profile configuration (not stored in DB)

```
maintenance-unit bgp default-bgp
  all-neighbor
maintenance-unit ospfv2 default-ospfv2
  all-process
```

```
maintenance-unit ospfv3 default-ospfv3
  all-process
maintenance-mode profile default-profile
  maintenance-unit apply default-bgp
  maintenance-unit apply default-ospfv2
  maintenance-unit apply default-ospfv3
```

Default Profile Feature-Sets

The default profile applies maintenance operations to the following protocols using protocol-specific maintenance-units:

1. **BGP - All neighbor sessions** (default-bgp maintenance-unit)
 - All BGP neighbor sessions across all VRFs are put into maintenance mode.
 - BGP graceful shutdown community is sent and sessions are brought down.
2. **OSPF - All processes** (default-ospfv2 and default-ospfv3 maintenance-units)
 - All OSPFv2 and OSPFv3 processes across all VRFs are put into maintenance mode using max-metric approach.
 - This gracefully removes the switch from routing.

Default Profile Recommendations

It is strongly recommended to create and configure a custom maintenance mode profile before activating maintenance mode. This approach provides several advantages:

- **Administrative Control:** Explicit control over which protocols and interfaces are affected
- **Selective Maintenance:** Ability to affect only specific network components rather than everything
- **Visibility:** Clear documentation of maintenance procedures in the device configuration
- **Predictability:** Consistent behavior across maintenance operations

Feature Interaction and Dependencies

Maintenance mode is designed to take precedence over other similar network features to ensure consistent and predictable behavior during maintenance operations.

Feature Precedence Behavior

When maintenance mode is active, it supersedes the following features:

- **OSPF max-metric commands:** Maintenance mode's max-metric implementation takes priority over manually configured max-metric settings
- **BGP neighbor shutdown commands:** Maintenance mode's graceful shutdown process overrides individual neighbor shutdown configurations

Conflict Avoidance Recommendations

It is strongly recommended to avoid configuring or modifying conflicting features while maintenance mode is active or configured. Specifically:

- Do not configure OSPF max-metric commands while maintenance mode profiles reference those OSPF processes.
- Do not issue BGP neighbor shutdown commands for neighbors included in maintenance mode maintenance-units.

These precautions prevent operational confusion and ensure that maintenance mode behavior remains predictable and consistent.

Configuration Change Guidelines

Maintenance mode requires configuration stability to ensure reliable and predictable operation throughout the maintenance window.

Feature-Set and Profile Modification

Feature-sets and profiles can be modified at any time. You can add or remove processes, neighbors, or other parameters within existing maintenance-units. However, changes made after maintenance mode is activated have indeterminate behavior and should be avoided.

Configuration Stability Requirements

It is critical not to perform configuration changes to maintenance mode profiles, maintenance-units, or referenced network protocols while maintenance mode is active. Configuration modifications during active maintenance can result in:

- **Inconsistent State:** Partial application of changes that may not align with the current maintenance stage
- **Unexpected Behavior:** Protocol behavior that differs from the configured maintenance profile
- **Traffic Disruption:** Unplanned traffic impact due to configuration timing conflicts
- **Operational Complexity:** Difficulty troubleshooting issues that arise from configuration changes during maintenance

Safe Configuration Practices

Follow these guidelines for safe maintenance mode operation:

1. **Pre-Maintenance Planning:** Complete all maintenance mode profile and maintenance-unit configurations before activating maintenance mode.
2. **Configuration Lock:** Treat maintenance mode activation as a configuration change freeze period.
3. **Emergency Changes:** If emergency configuration changes are absolutely necessary, deactivate maintenance mode first, make the changes, then reactivate if needed.

4. **Post-Maintenance Updates:** Schedule any routine configuration changes for after maintenance mode deactivation and normal operation verification.

Recommended Workflow

1. Plan maintenance requirements and timeline
2. Configure maintenance-units and maintenance mode profile
3. Verify configuration with show commands
4. Activate maintenance mode (configuration freeze begins)
5. Perform physical maintenance tasks
6. Deactivate maintenance mode
7. Verify normal operation restoration
8. Resume normal configuration management activities

This workflow ensures that maintenance mode operates in a stable configuration environment, maximizing the reliability and predictability of the maintenance process.

VSX/MCLAG-Specific Behavior and Scope

What Maintenance Mode does not Include for VSX

- **Important Distinction:** Maintenance mode on a VSX peer is not equivalent to VSX software upgrade procedures or comprehensive VSX maintenance workflows.
- VSX-Specific Operations NOT Handled by Maintenance Mode.

Maintenance mode does not perform the following VSX-specific operations:

- **VSX ISL (Inter-Switch Link) Maintenance:** Maintenance mode does not affect VSX ISL link states or perform ISL-specific maintenance procedures.
- **VSX Keepalive Management:** VSX keepalive link state and keepalive protocol operation are not modified by maintenance mode.
- **VSX Split-Brain Prevention:** Maintenance mode does not alter VSX split-brain detection or prevention mechanisms.
- **VSX Configuration Synchronization:** Maintenance mode configurations (maintenance-units, profiles, and activation state) are ****not synchronized**** via VSX config-sync between VSX peers. Each VSX peer must have its maintenance mode configuration independently configured. Config-sync pause or control is not part of maintenance mode operations.

- **VSX Role Sequencing:** Maintenance mode does not enforce or recommend specific VSX primary/secondary role sequencing for maintenance.
- **VSX Device Parameter Management:** VSX device-specific parameters (system-mac, isl-port, keepalive-peer, device-role) are not affected.
- **VSX Software Upgrade Distinction:** Maintenance mode is not a substitute for VSX software upgrade procedures. VSX software upgrades have specific workflows, sequencing requirements, and safety mechanisms that are separate from maintenance mode protocol signaling.

When to use Maintenance Mode on VSX Peers

Selective Protocol Maintenance: Maintenance targeting specific OSPF processes or BGP neighbors.

VSX Maintenance Mode Behavior Details

Traffic redirection in VSX environments:

- OSPF max-metric causes routing traffic to prefer the non-maintenance VSX peer if available.
- BGP graceful shutdown redirects BGP traffic to alternate paths, potentially including the other VSX peer.

Maintenance mode on VSX peers provides protocol-level traffic redirection for BGP and OSPF but does not replace VSX-specific maintenance or upgrade procedures.

High Availability (HA) Behavior

Maintenance mode has specific behaviors and restrictions in High Availability (HA) configurations that network operators must understand when planning maintenance.

Activation Restrictions

- Active Switch Only: Maintenance mode can only be activated on the **active** switch in an HA pair. The standby switch does not accept maintenance mode activation commands.
- Command Behavior:
 - On Active Switch: **maintenance-mode activate** command executes successfully and puts the active switch into maintenance mode.
 - On Standby Switch: **maintenance-mode activate** command returns an error indicating that maintenance mode cannot be activated on standby.

This restriction ensures that:

1. Only the switch actively forwarding traffic can signal maintenance state to the network.
2. Configuration changes that affect active traffic flows are controlled from the active switch.

3. Standby switch remains in ready state to assume active role if needed.

Failover Behavior

Ephemeral State: Maintenance mode state is not maintained across HA failover events. When an HA failover occurs:

1. Previous Active Switch: Maintenance mode state is lost when the switch transitions to standby role.
2. New Active Switch: Assumes active role in normal operation mode (not in maintenance mode).
3. Network Operation: Network protocols on the new active switch begin normal operation immediately.

Post-Failover Actions: After an HA failover, if maintenance operations still need to continue:

1. Verify that the new active switch has stabilized and is forwarding traffic normally.
2. If maintenance mode is still needed, run **maintenance-mode activate [profile <PROFILE_NAME>]** on the new active switch.
3. All maintenance mode configurations (profiles and maintenance-units) remain in the device configuration and can be reactivated.

HA Maintenance Planning

Recommended Approach for maintenance in HA environments:

1. Pre-Maintenance Verification:
 - Verify HA status and identify the active switch
 - Ensure HA synchronization is complete
 - Verify both switches are healthy
2. Maintenance Execution:
 - Activate maintenance mode on the active switch only
 - Perform maintenance tasks on the active switch
 - Monitor HA status throughout maintenance
3. Failover Scenarios:
 - If unplanned failover occurs during maintenance, maintenance mode is automatically cleared
 - Assess whether to reactivate maintenance mode on the new active switch
 - Consider completing maintenance on the original switch after it returns to service
4. Planned Failover:
 - Deactivate maintenance mode before initiating planned failover

- Allow network to stabilize in normal operation
- If needed, activate maintenance mode on the new active switch after failover completes

Configuration Synchronization:

Maintenance mode profiles and maintenance-units are synchronized between HA peers as part of normal configuration synchronization. However, the active/inactive state of maintenance mode itself is not synchronized and remains local to the active switch.

Known Issues and Limitations

This section documents current limitations and known issues with the maintenance mode implementation that network operators should be aware of when planning maintenance operations.

BGP Address Family Limitations

Supported Address Families	Maintenance mode BGP maintenance-units currently support only the following address families: • IPv4 Unicast: Fully supported for graceful shutdown and maintenance operations. • IPv6 Unicast: Fully supported for graceful shutdown and maintenance operations.
Unsupported Address Families	The following BGP address families are not qualified for maintenance mode operations: • EVPN Route-Type 2 (MAC/IP Advertisement Routes): EVPN MAC/IP routes are not included in maintenance mode graceful shutdown procedures. • EVPN Route-Type 5 (IP Prefix Routes): EVPN IP prefix routes are not included in maintenance mode graceful shutdown procedures.
Impact	BGP neighbors configured with EVPN address families will not receive graceful shutdown treatment for EVPN routes during maintenance mode activation. Only IPv4 and IPv6 unicast routes will be gracefully withdrawn.
Recommended Workaround	For networks heavily dependent on EVPN services, consider: 1. Manual EVPN neighbor shutdown before activating maintenance mode. 2. Coordinating maintenance windows with EVPN convergence timelines. 3. Monitoring EVPN route convergence separately from maintenance mode operations.

OSPF Standards Compliance Limitations

RFC 8379 not supported	Maintenance mode OSPF implementation does not include support for RFC 8379 (OSPF Graceful Link Shutdown). Current Implementation: The maintenance mode OSPF maintenance-units use the traditional max-metric approach rather than the newer graceful link shutdown mechanisms defined in RFC 8379.
Impact	Networks expecting RFC 8379 compliant graceful link shutdown behavior will experience traditional max-metric behavior instead.
Current Behavior	• OSPF processes use maximum metric values (0xFFFF) in LSA advertisements. • Standard OSPF convergence procedures apply. • No RFC 8379 specific graceful shutdown signaling.
Planning Considerations	1. Factor traditional OSPF convergence times into maintenance windows. 2. Monitor OSPF neighbor states and LSA propagation during maintenance operations.

Supportability and Diagnostics Limitations

SAINT Integration	SAINT (Service Availability Insight) is not supported in this release.
-------------------	--

General Recommendations

Given these limitations, network operators should:

1. **Pre-Maintenance Testing:** Validate maintenance mode behavior in lab environments that match production network characteristics.
2. **Selective Application**
: Consider creating maintenance-units that exclude problematic address families or routing protocols.
3. **Manual Coordination:**
: Plan manual intervention for unsupported protocol features during maintenance windows.
4. **Monitoring Strategy:**

Implement comprehensive monitoring for both supported and unsupported protocol behaviors during maintenance.

These limitations are documented to ensure proper maintenance planning and to set appropriate expectations for maintenance mode behavior in complex network environments.

Maintenance Mode Scope and Traffic Impact Analysis

This section provides a comprehensive analysis of what maintenance mode addresses in the current release and clearly identifies scenarios that are out of scope. This compartmentalized view helps network operators understand the specific traffic impact of maintenance mode operations.

Current Release Scope

Maintenance mode in this release is specifically designed to address *3 Unicast dynamic routing protocols:

Supported Protocols	Traffic Types
OSPF (OSPFv2, OSPFv3, OSPFv3-AF)	IPv4 and IPv6 unicast routing via OSPF max-metric
BGP IPv4 Unicast	IPv4 unicast prefix advertisement and neighbor session management
BGP IPv6 Unicast	IPv6 unicast prefix advertisement and neighbor session management

Traffic Impact: When maintenance mode is activated, these protocols gracefully redirect L3 unicast routing traffic away from the switch undergoing maintenance, minimizing disruption to unicast IP forwarding.

L3 Unicast Out-of-Scope Features

The following L3 unicast routing features and scenarios are not addressed by maintenance mode in the current release:

Routing Protocols and Features	Description
Static Routes	Maintenance mode does not modify or disable static route configurations
Policy-Based Routing (PBR)	PBR configurations remain active and unchanged during maintenance
VRRP	First-hop redundancy protocol is not affected by maintenance mode

Routing Protocols and Features	Description
Route Maps and Filters	Existing route manipulation policies continue to operate normally.
Multicast Routing Protocols	PIM, IGMP, and MLD are not in scope (see L2/L3 Multicast section below).
BGP Advanced Features:	
BGP EVPN Address Family	EVPN Route-Type 2 (MAC/IP) and Route-Type 5 (IP Prefix) routes are not qualified for maintenance mode.
BGP VPNv4/VPNv6	MPLS L3VPN address families are not in scope.
BGP Multicast Address Family	BGP multicast prefixes are not affected.

These out-of-scope L3 features continue normal operation during maintenance mode. Static routes, PBR, and VRRP will continue to forward traffic, potentially preventing complete traffic redirection if they provide active paths to destinations.

L2 Unicast Out-of-Scope Features

All L2 unicast forwarding mechanisms and protocols are out of scope for maintenance mode in the current release:

L2 Protocols and Features	Description
Spanning Tree Protocol (STP/RSTP/MSTP)	STP states and bridge priorities are not modified.
Link Aggregation (LAG/LACP)	Interface Draining feature is currently out of scope.
MAC Address Learning	Dynamic MAC learning continues normally.
VLAN Forwarding	Layer 2 VLAN forwarding behavior is unchanged.
Multicast Routing Protocols	PIM, IGMP, and MLD are not in scope (see L2/L3 Multicast section below).

L2 unicast traffic forwarding continues normally during maintenance mode. Switches remain active participants in STP topology, LAG bundles remain operational, and MAC learning continues. This means L2-switched traffic is not redirected during maintenance operations. LACP/LAG maintenance capabilities are planned in a future release, which will provide graceful L2 link shutdown mechanisms.

L2/L3 Multicast Out-of-Scope Features

All multicast routing and forwarding features are out of scope for maintenance mode in the current release:

L2 Protocols and Features	Description
PIM (Protocol Independent Multicast)	PIM-SM, PIM-DM, and PIM-SSM operations are not affected.
IGMP (Internet Group Management Protocol)	IGMP snooping and IGMP querier functions continue normally.
MAC Address Learning	MLD for IPv6 multicast operates unchanged.
MLD (Multicast Listener Discovery)	Layer 2 VLAN forwarding behavior is unchanged.
Multicast VPN	L3 multicast VPN services are not in scope.
MVR (Multicast VLAN Registration):	MVR configurations remain active.

Multicast traffic (both L2 and L3) continues to flow normally through the switch during maintenance mode. The switch remains an active participant in multicast distribution trees, continues to process IGMP/MLD reports, and forwards multicast traffic based on existing state. Multicast maintenance capabilities are under consideration for future releases, which will provide graceful multicast traffic redirection mechanisms.

Key Principle

Maintenance mode is narrowly scoped to L3 unicast dynamic routing protocols (OSPF and BGP unicast) only.

All other traffic types, forwarding mechanisms, and protocols continue normal operation during maintenance. Network operators must plan maintenance procedures accounting for traffic that is not redirected by maintenance mode, including L2 switched traffic, multicast traffic, and traffic forwarded via static routes or other out-of-scope mechanisms.

For comprehensive traffic redirection during physical switch maintenance, operators may need to combine maintenance mode activation with additional manual procedures for out-of-scope features.

Show Events

Monitor the progress of maintenance mode activation, deactivation and diagnose graceful shutdown sequencing for BGP and OSPF protocols. The **show events -d gshut-daemon** command displays the event logs generated by **gshut-daemon**:

Event Name	Description
GSHUT_PROFILE_ACTIVATED	Logs event when a maintenance mode profile is activated
GSHUT_PROFILE_DEACTIVATED	Logs event when a maintenance mode profile is deactivated

Event Name	Description
GSHUT_FEATURE_SET_STATUS_CHANGE	Logs event when a routing process completes its graceful shutdown during maintenance
GSHUT_BGP_COORDINATION_CHANGE	Logs event when BGP graceful shutdown progresses to a new phase
GSHUT_BGP_TIMER_STARTED	Logs event when BGP graceful shutdown timer starts
GSHUT_BGP_TIMER_EXPIRED	Logs event when BGP graceful shutdown timer expires and triggers neighbor session shutdown

REST API Terminology and Data Model Reference

This section provides comprehensive mapping between CLI terminology and REST API data structures for maintenance mode configuration. This reference is intended for REST API developers and automation engineers who need to understand how CLI concepts translate to REST API operations.

CLI to REST API Terminology Mapping

CLI Terminology	REST API Schema	Yang Path	Description
maintenance-unit bgp	Feature_Set_BGP	<code>/system/feature_set_bgps</code>	BGP-specific maintenance configurations
maintenance-unit ospfv2 /ospfv3/ospfv3-af	Feature_Set_OSPF	<code>/system/feature_set_ospfs</code>	OSPF-specific maintenance configurations (all variants share one table)
maintenance-mode profile	Maintenance_Mode_Profile	<code>/system/maintenance_mode_profiles</code>	Top-level orchestration linking maintenance-units to stages
Not visible in CLI	Maintenance_Stage	<code>/system/maintenance_mode_profiles/stages</code>	System-managed container grouping maintenance-units

Key Architecture Change:

The REST API uses two separate protocol-specific tables (**Feature_Set_BGP**, **Feature_Set_OSPF**) instead of a single multi-protocol table. This aligns with the CLI's protocol-specific maintenance-unit model. Stages link to feature-sets via URI references in separate typed arrays.

Data Model Hierarchy

Maintenance_Mode_Profile (Top Level)

name: "campus-mode-profile"

activated: true/false (controls maintenance mode state)

stages: {1: <stage-uri>} (map: stage_id → Stage)

Stage (stage_id=1, created via POST)

feature_set_bgp:
["/rest/latest/system/feature_set_bgps/campus-bgp"]
(array of URI references to Feature_Set_BGP resources)

feature_set_ospf:
["/rest/latest/system/feature_set_ospfs/campus-ospfv2"]
(array of URI references to Feature_Set OSPF resources)

stage_delay: 0-300 seconds

Feature_Set_BGP "campus-bgp"

bgp_peer_vrfs: {peer_address: "/rest/latest/system/vrfs/<name>"}

shutdown_delay_timer: 120

all_instances: true/false

Feature_Set OSPF "campus-ospfv2"

protocol_type: "ospfv2" | "ospfv3" | "ospfv3-af-ipv4" | "ospfv3-af-ipv6"

ospf_process_vrfs: {process_id: "/rest/latest/system/vrfs/<name>"}

ospf_all_vrfs_process_ids: [1, 2, 3]

The REST API uses a 3-tier hierarchy with protocol-specific tables:

Subtopics

Key REST API Concepts

Protocol-Specific Tables

By design, feature-sets are stored in two separate protocol-specific tables:

Feature-set	Table	Fields
Feature_Set_BGP	/rest/latest/system/feature_set_bgps Contains only BGP-related fields	bgp_peer_vrfs shutdown_delay_timer all_instances
Feature_Set_OSPF	/rest/latest/system/feature_set_ospfs Handles ALL OSPF variants (OSPFv2, OSPFv3, OSPFv3-AF-IPv4, OSPFv3-AF-IPv6)	protocol_type ospf_process_vrfs ospf_all_vrfs_process_ids all_instances

Typed Feature-Set Arrays in Stages

Stages use separate typed arrays with URI references for each protocol:

```
{
  "feature_set_bgp": ["/rest/latest/system/feature_set_bgps/campus-bgp"],
  "feature_set_ospf": [
    "/rest/latest/system/feature_set_ospfs/campus-ospfv2",
    "/rest/latest/system/feature_set_ospfs/campus-ospfv3"
  ]
}
```

Each protocol has its own array, using URI references (not UUIDs) to identify feature-sets.

VRF Name to URI Reference Resolution

CLI uses **VRF names**, but REST API uses **VRF URI references**:

```
curl -X PATCH "/rest/latest/system/feature_set_ospfs/campus-ospfv2" \
-H "Content-Type: application/json" \
-d '{"ospf_process_vrfs": {"1": "/rest/latest/system/vrfs/red"}}'
```

Stage Creation:

When you create a **Maintenance_Mode_Profile**, stages start empty. You must explicitly create stages via POST to link feature-sets:

Create mode profile (starts with empty stages):

```
curl -X POST "/rest/latest/system/maintenance_mode_profiles" \
-H "Content-Type: application/json" \
-d '{"name": "campus-mode-profile"}'
```

Create stage and link feature-sets:

```
curl -X POST "/rest/latest/system/maintenance_mode_profiles/campus-mode-profile/stages" \
-H "Content-Type: application/json" \
-d '{
  "stage_id": 1,
  "feature_set_bgp": ["/rest/latest/system/feature_set_bgps/campus-bgp"],
  "feature_set_ospf": ["/rest/latest/system/feature_set_ospfs/campus-ospfv2"]
}'
```

Activation Control

Maintenance mode is activated or deactivated by setting the **activated** boolean on the **Maintenance_Mode_Profile**:

Activate maintenance mode:

```
curl -X PATCH "/rest/latest/system/maintenance_mode_profiles/campus-mode-profile" \
-H "Content-Type: application/json" \
-d '{"activated": true}'
```

Deactivate maintenance mode:

```
curl -X PATCH "/rest/latest/system/maintenance_mode_profiles/campus-mode-profile" \
-H "Content-Type: application/json" \
-d '{"activated": false}'
```

REST API Base Paths

All maintenance mode REST APIs use the following base path:

```
/rest/latest/system/feature_set_bgps
(Feature_Set_BGP)
/rest/latest/system/feature_set_ospfs
(Feature_Set OSPF)
/rest/latest/system/maintenance_mode_profiles
(Maintenance_Mode_Profile)
```



```
/rest/latest/system/maintenance_mode_profiles/{name}/stages
(Maintenance_Stage - POST to create)
/rest/latest/system/maintenance_mode_profiles/{name}/stages/{stage_id}
(GET/DELETE)
```

Complete REST API Workflow Example

1. Create BGP maintenance-unit:

```
curl -X POST "/rest/latest/system/feature_set_bgps" \
-H "Content-Type: application/json" \
-d '{
  "name": "campus-bgp",
  "bgp_peer_vrfs": {"10.1.1.1": "/rest/latest/system/vrfs/red",
"10.1.1.2": "/rest/latest/system/vrfs/red"},
  "shutdown_delay_timer": 120
}'
```

2. Create OSPF maintenance-unit:

```
curl -X POST "/rest/latest/system/feature_set_ospfs" \
-H "Content-Type: application/json" \
-d '{
  "name": "campus-ospfv2",
  "protocol_type": "ospfv2",
  "ospf_process_vrfs": {"1": "/rest/latest/system/vrfs/red", "2": "/rest/
latest/system/vrfs/red"}
}'
```

3. Create mode profile:

```
curl -X POST "/rest/latest/system/maintenance_mode_profiles" \
-H "Content-Type: application/json" \
-d '{"name": "campus-mode-profile"}'
```

4. Create stage and link feature-sets via URI references:

```
curl -X POST "/rest/latest/system/maintenance_mode_profiles/campus-mode-
profile/stages" \
-H "Content-Type: application/json" \
-d '{
  "stage_id": 1,
  "feature_set_bgp": ["/rest/latest/system/feature_set_bgps/campus-bgp"],
  "feature_set_ospf": ["/rest/latest/system/feature_set_ospfs/campus-
ospfv2"]
}'
```

5. Activate maintenance mode:

```
curl -X PATCH "/rest/latest/system/maintenance_mode_profiles/campus-mode-
profile" \
```

```
-H "Content-Type: application/json" \  
-d '{"activated": true}'
```

6. Check status:

```
curl -X GET "/rest/latest/system/maintenance_mode_profiles/campus-mode-  
profile"
```

7. Deactivate when complete

```
curl -X PATCH "/rest/latest/system/maintenance_mode_profiles/campus-mode-  
profile" \  
-H "Content-Type: application/json" \  
-d '{"activated": false}'
```



NOTE

REST API Usage Notes for REST API developers:

- Protocol-specific maintenance-units are stored in two separate tables
 - Stages use separate **feature_set_bgp** and **feature_set_ospf** arrays with URI references
 - VRF references use URI paths directly, for example **/rest/latest/system/vrfs/red**) — no UUID lookup needed
 - Stages must be explicitly created via POST after creating the mode profile
 - PATCH on stages is additive only; use DELETE + POST to replace stage content
 - Activation state is controlled at the **Maintenance_Mode_Profile** level, not individual maintenance-units
- One **Feature_Set_OSPF** table handles all OSPF variants (v2, v3, v3-AF-IPv4, v3-AF-IPv6)
- OSPF POST requires **protocol_type** field to specify the variant

Maintenance mode commands

Subtopics

maintenance-mode activate [profile <PROFILE_NAME>]

maintenance-mode profile <PROFILE_NAME>

maintenance-unit apply <FEATURE_SET_NAME>

maintenance-unit bgp <FEATURE_SET_NAME>

maintenance-unit ospfv2 <FEATURE_SET_NAME>
maintenance-unit ospfv3 <FEATURE_SET_NAME>
maintenance-unit ospfv3-af-ipv4 <FEATURE_SET_NAME>
maintenance-unit ospfv3-af-ipv6 <FEATURE_SET_NAME>
neighbor <IP_ADDRESS> vrf <VRF_NAME>
process <PROCESS_ID>
show events -d gshut-daemon
show maintenance-mode profile
show maintenance-mode
show maintenance-unit
shutdown-delay-timer <SECONDS>

maintenance-mode activate [profile <PROFILE_NAME>]

Syntax

```
[no] maintenance-mode activate [profile <PROFILE_NAME>]
```

Description

Activates maintenance mode on the Aruba switch. This command puts the switch into maintenance mode, allowing network operators to perform maintenance tasks while minimizing disruption to live network traffic. The command uses protocol mechanisms (OSPF max-metric, BGP GSHUT community with shutdown timer) to gracefully reroute traffic away from the switch.

The **no** form of this command seactivates maintenance mode on the Aruba switch. This command takes the switch out of maintenance mode and restores normal operation, regardless of whether maintenance mode was activated with the default profile or a custom profile. All maintenance mode configurations including profiles and feature-sets remain configured and can be reactivated later.



NOTE

Deactivation can cause traffic loss during BGP neighbor reconvergence. BGP sessions that were shut down during maintenance mode must re-establish, re-exchange routes, and update routing tables. This reconvergence process takes time (seconds to minutes) and may disrupt traffic. Plan deactivation during appropriate maintenance windows. See the "Design Principles and Traffic Impact" section for detailed information.

Parameter	Description
activate	Activates maintenance mode state.

Parameter	Description
	Required
<code>profile <PROFILE_NAME></code>	Name of the pre-configured profile.
	Optional

Usage

Traffic Impact Warning: Activation minimizes but does not eliminate traffic loss. Brief disruption may occur during protocol reconvergence to alternate paths. Redundant network paths are required for proper operation. See the Design Principles and Traffic Impact section for detailed information.

Default Profile Activation: When executed without the profile parameter, **maintenance-mode activate** uses the built-in default profile which affects all OSPF processes, all BGP neighbors, and all VSX/MCLAG LAG ports across all VRFs.

Custom Profile Activation: When executed with **profile <PROFILE_NAME>**, maintenance mode uses the specified custom profile. The profile must be pre-configured. If the specified profile does not exist, the command returns an error.

High Availability (HA) Restriction: Maintenance mode can only be activated on the active switch in an HA configuration. Attempting to activate maintenance mode on the standby switch will result in an error.

Configuration Freeze During Activation: Once maintenance mode is activated, any modifications to maintenance-units (BGP or OSPF) or custom maintenance-mode profiles will not be applied dynamically at runtime. Changes to maintenance-unit configurations or profile associations made after activation will only take effect after deactivating and reactivating maintenance mode. To apply updated configurations, deactivate maintenance mode first using **no maintenance-mode activate**, make the necessary changes, and then reactivate maintenance mode.

Maintenance mode is active until manually deactivated or system reboot. After reboot or HA failover, the system returns to normal operation.

Examples

Activating maintenance mode with default profile:

```
switch(config) # maintenance-mode activate
```

Activating maintenance mode with a custom profile:

```
switch(config) # maintenance-mode activate profile campus-profile
```

Activating maintenance mode when no profile is configured (shows default behavior note):

```
switch(config) # maintenance-mode activate
```

```
Note: No maintenance profile configured, using default behavior (all OSPF,
BGP)
```

Activating maintenance mode with a profile that has no stages configured (shows warning):

```
switch(config)# maintenance-mode activate profile empty-profile  
Warning: Maintenance mode profile has no stages configured.  
Using default behavior (all OSPF, BGP).
```

Attempting to activate when maintenance mode is already active with same profile:

```
switch(config)# maintenance-mode activate profile campus-profile  
Maintenance mode is already activated
```

Attempting to activate when maintenance mode is already active with different profile

```
switch(config)# maintenance-mode activate profile other-profile  
Error: Maintenance mode is already activated for profile 'campus-profile'.  
Only one profile can be activated at a time.  
Please deactivate the current profile first using 'no maintenance-mode  
activate'
```

Activating with non-existent profile (error)

```
switch(config)# maintenance-mode activate profile nonexistent  
Profile 'nonexistent' not found
```

Deactivating maintenance mode :

```
switch(config)# no maintenance-mode activate
```

REST API Examples

Activating maintenance mode on node profile:

```
curl -X PATCH "/rest/latest/system/maintenance_mode_profiles/campus-mode-  
profile" \  
-H "Content-Type: application/json" \  
-d '{"activated": true}'
```

Deactivating maintenance mode:

```
curl -X PATCH "/rest/latest/system/maintenance_mode_profiles/campus-mode-  
profile" \  
-H "Content-Type: application/json" \  
-d '{"activated": false}'
```

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config	Administrators or local user group members with execution rights for this command.
6400		
8100		
8325		
8325H		
8360		
8400		
9300		
9300-32D		
10000		

maintenance-mode profile <PROFILE_NAME>

Syntax

```
[no] maintenance-mode profile <PROFILE_NAME>
```

Description

Creates or enters into the maintenance mode profile context. This command allows you to define a custom maintenance mode profile that references pre-created BGP and OSPF maintenance-units. The profile can reference one or multiple maintenance-units depending on maintenance requirements. To use a configured profile, it must be explicitly specified in the **maintenance-mode activate profile <PROFILE_NAME>** command.

The **no** form of this command deletes a maintenance-mode profile from the configuration. This command removes the entire maintenance-mode profile including all maintenance-unit application references. The profile cannot be deleted while maintenance mode is active using that profile. You must first deactivate maintenance mode using **no maintenance-mode activate** before deleting the profile. Maintenance-units referenced by the profile remain configured and can be applied to other profiles. Only one maintenance-

mode profile can exist at a time, so deleting the current profile allows the creation of a different profile configuration.



NOTE

Only one custom maintenance-mode profile can be configured at a time in this release. Attempting to create a second profile while one already exists will be rejected by the system. However, multiple maintenance-units can coexist and be reused. To create a different maintenance-mode profile configuration, you must first delete the existing maintenance-mode profile using **no maintenance-mode profile <PROFILE_NAME>**, then create the new maintenance-mode profile. Feature-sets remain configured and can be reused when creating different maintenance-mode profile configurations.

Parameter	Description
<PROFILE_NAME>	Name of the maintenance mode profile. Required

Examples

Creating and entering maintenance mode profile named **campus-profile**

```
switch(config)# maintenance-mode profile campus-profile
switch(config-maint-mode-campus-profile)#
```

In this example, a custom maintenance-mode profile already exists and another profile is created. This will be rejected.

```
switch(config)# maintenance-mode profile ospf
switch(config-maint-mode-ospf)# exit
switch(config)# maintenance-mode profile ospf1
% Only one custom maintenance-mode profile is allowed, delete existing one
to create a new maintenance-mode profile.
```

To resolve this, delete the existing profile using **no maintenance-mode profile <EXISTING_PROFILE_NAME>** before creating a new one.

Deleting maintenance-mode profile named **campus-profile**

```
switch(config)# no maintenance-mode profile campus-profile
```

Deleting an active profile will fail:

```
switch(config)# no maintenance-mode profile campus-profile
Error: Cannot delete profile 'campus-profile' - maintenance mode is
currently active
```

To achieve this deactivate first, then delete:

```
switch(config) # no maintenance-mode activate
switch(config) # no maintenance-mode profile campus-profile
```

REST API Examples

Creating maintenance mode profile (stages start empty; create via POST separately)

```
curl -X POST "/rest/latest/system/maintenance_mode_profiles" \
-H "Content-Type: application/json" \
-d '{
  "name": "campus-mode-profile",
  "description": "Campus network maintenance orchestration"
}'
```

Delete maintenance node profile (CLI **maintenance-mode profile** maps to **Maintenance_Mode_Profile** in REST):

```
curl -X DELETE "/rest/latest/system/maintenance_mode_profiles/campus-mode-profile"
```

REST API will fail if node profile is currently activated. First deactivate, then delete:

```
curl -X DELETE "/rest/latest/system/maintenance_mode_profiles/campus-mode-profile"
-H "Content-Type: application/json" \
-d '{"activated": false}'
curl -X DELETE "/rest/latest/system/maintenance_mode_profiles/campus-mode-profile"
```

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config	Administrators or local user group members with execution rights for this command.
6400	config-feature-set-bgp-campus-bgp	
8100		
8325		
8325H		
8360		

Platforms	Command context	Authority
-----------	-----------------	-----------

8400

9300

9300-32D

10000

maintenance-unit apply <FEATURE_SET_NAME>

Syntax

```
[no] maintenance-unit apply <FEATURE_SET_NAME>
```

Description

Applies a pre-created maintenance-unit within a maintenance mode profile. This command references a protocol-specific maintenance-unit that was previously created at the global configuration level and applies it to the profile. Each feature-set is dedicated to a single protocol (BGP or OSPF), and multiple maintenance-units can be applied within the same profile to affect multiple protocols during maintenance.

The **no** form removes a maintenance-unit application from a maintenance-mode profile. This command unlinks the specified maintenance-unit from the profile without deleting the maintenance-unit itself. The maintenance-unit remains configured at the global level and can be applied to other profiles or reapplied later. This removal can be performed regardless of whether maintenance mode is active, but will affect maintenance operations if performed while the profile is active. If you remove all maintenance-units from a profile, the profile becomes empty but still exists.

Parameter

Description

<FEATURE_SET_NAME>

BGP neighbor IP address (when using specific neighbor).

Examples

Applying BGP maintenance-unit within a profile:

```
switch(config)# maintenance-mode profile campus-profile
switch(config-maint-mode-campus-profile)# maintenance-mode apply campus-bgp
```

Applying multiple protocol-specific maintenance-units within a profile:

```
switch(config-maint-mode-campus-profile)# maintenance-mode apply campus-bgp
switch(config-maint-mode-campus-profile)# maintenance-mode apply campus-
ospfv2
```

Removing BGP maintenance-unit from a profile:

```
switch(config)# maintenance-mode profile campus-profile
switch(config-maint-mode-campus-profile)# no maintenance-unit apply campus-
bgp
```

Removing one of multiple maintenance-units from a profile:

```
switch(config)# maintenance-mode profile campus-profile
switch(config-maint-mode-campus-profile)# no maintenance-mode apply
underlay-ospfv2
```

REST API Examples

Linking maintenance-units to mode profile by creating a stage with feature-set references. Custom profiles start with empty stages; create stage via POST with feature-set URI references

```
curl -X POST "/rest/latest/system/maintenance_mode_profiles/campus-mode-
profile/stages" \
-H "Content-Type: application/json" \
-d '{
  "stage_id": 1,
  "feature_set_bgp": ["/rest/latest/system/feature_set_bgps/campus-bgp"],
  "feature_set_ospf": ["/rest/latest/system/feature_set_ospfs/campus-
ospfv2"]
}'
```

Removing maintenance-unit from profile by replacing stage content:



NOTE

PATCH on stages is additive only (cannot remove feature-sets). Use DELETE + POST to replace.

Delete existing stage and recreate with only the desired feature-sets.

```
curl -X DELETE "/rest/latest/system/maintenance_mode_profiles/campus-mode-
profile/stages/1"
```

Recreate stage with only campus-ospfv2 (campus-bgp removed):

```
curl -X POST "/rest/latest/system/maintenance_mode_profiles/campus-mode-
profile/stages" \
-H "Content-Type: application/json" \
-d '{
  "stage_id": 1,
  "feature_set_ospf": ["/rest/latest/system/feature_set_ospfs/campus-
ospfv2"]
}'
```

Remove all maintenance-units from profile (delete stage entirely)

```
curl -X DELETE "/rest/latest/system/maintenance_mode_profiles/campus-mode-
profile/stages/1"
```

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config	Administrators or local user group members with execution rights for this command.
6400		
8100		
8325		
8325H		
8360		
8400		
9300		
9300-32D		
10000		

`maintenance-unit bgp <FEATURE_SET_NAME>`

Syntax

```
[no] maintenance-unit bgp <FEATURE_SET_NAME>
```

Description

Creates and enters a BGP maintenance-unit context at the global configuration level. This command allows you to define reusable BGP- specific maintenance-units for maintenance mode operations. BGP maintenance-units configure graceful shutdown behavior for BGP neighbor sessions including shutdown delay timers and neighbor selection. Once created, you can configure BGP- specific parameters directly (no sub-context). These BGPmaintenance-units can be referenced and applied within maintenance mode profiles.

The **no** form of this command deletes a BGP maintenance-unit from the global configuration. It also removes the entire BGP maintenance-unit including all BGP neighbor and timer configurations. The BGP

maintenance-unit cannot be deleted if it is currently applied to a maintenance-mode profile. You must first remove the maintenance-unit from any profiles using **no maintenance-unit <FEATURE_SET_NAME>** before deletion. The maintenance-unit can be deleted regardless of whether maintenance mode is active, as long as it is not applied to the active profile.

Parameter	Description
<FEATURE_SET_NAME>	Name of the BGP maintenance-unit. Required

Examples

Creating BGP maintenance-unit named **campus-bgp** with neighbor configuration:

```
switch(config) # maintenance-unit bgp campus-bgp
switch(config-maintenance-unit-bgp-campus-bgp) # neighbor 10.1.1.1 vrf red
```

Creating a BGP maintenance-unit with all-neighbor configuration:

```
switch(config) # maintenance-unit bgp campus-bgp
switch(config-maintenance-unit-bgp-campus-bgp) # all-neighbor
```

Deleting a BGP maintenance-unit named **campus-bgp**:

```
switch(config) # no maintenance-unit bgp campus-bgp
```

Attempting to delete BGP maintenance-unit that is applied to a profile. This will fail:

```
switch(config) # no maintenance-unit bgp campus-bgp
Error: Cannot delete maintenance-unit 'campus-bgp' - currently applied to
maintenance-mode profile 'campus-profile'
```

The correct sequence is to remove from profile first, then delete the maintenance-unit:

```
switch(config) # maintenance-mode profile campus-profile
switch(config-maint-mode-campus-profile) # no maintenance-unit apply campus-
bgp
switch(config-maint-mode-campus-profile) # exit
switch(config) # no maintenance-unit bgp campus-bgp
```

REST API Examples

Creating a BGP maintenance-unit with specific neighbor:

(CLI **maintenance-unit bgp** maps to **Feature_Set_BGP** table in REST)

```
curl -X POST "/rest/latest/system/feature_set_bgps" \
-H "Content-Type: application/json" \
-d '{
  "name": "campus-bgp",
  "bgp_peer_vrfs": {"10.1.1.1": "/rest/latest/system/vrfs/red"},
}
```

```
"shutdown_delay_timer": 120
}'
```

Creating a BGP maintenance-unit with all neighbors:

```
curl -X POST "/rest/latest/system/feature_set_bgps" \
-H "Content-Type: application/json" \
-d '{
  "name": "campus-bgp",
  "all_instances": true,
  "shutdown_delay_timer": 120
}'
```

Attempting to delete BGP maintenance-unit that is applied to a profile. REST API will fail if maintenance-unit is referenced by any node profile stage. First, remove the reference from the stage:

```
curl -X DELETE "/rest/latest/system/maintenance_mode_profiles/campus-mode-
profile/stages/1"
```

Then delete the maintenance-unit:

```
curl -X DELETE "/rest/latest/system/feature_set_bgps/campus-bgp"
```

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config	Administrators or local user group members with execution rights for this command.
6400		
8100		
8325		
8325H		
8360		
8400		
9300		
9300-32D		
10000		

maintenance-unit ospfv2 <FEATURE_SET_NAME>

Syntax

```
[no] maintenance-unit ospfv2 <FEATURE_SET_NAME>
```

Description

Creates and enters an OSPFv2 maintenance-unit context at the global configuration level. This command allows you to define reusable OSPFv2-specific maintenance-unit for maintenance mode operations. OSPFv2 maintenance-unit configure max-metric behavior for OSPFv2 processes supporting IPv4 routing. Once created, you can configure OSPFv2 process IDs and VRFs directly (no sub-context). These OSPFv2 maintenance-unit can be referenced and applied within maintenance mode profiles.

The **no** form of this command deletes an OSPF maintenance-unit from the global configuration. It also removes the entire OSPF maintenance-unit including all OSPF process and VRF configurations. The OSPF maintenance-unit cannot be deleted if it is currently applied to a maintenance-mode profile.



NOTE

You must first remove the maintenance-unit from any profiles using **no maintenance-unit apply <FEATURE_SET_NAME>** before deletion. The maintenance-unit can be deleted regardless of whether maintenance mode is active, as long as it is not applied to the active profile.

Parameter	Description
<FEATURE_SET_NAME>	Name of the OSPFv2 maintenance-unit. Required

Examples

Creating OSPFv2 maintenance-unit named **campus-ospfv2**:

```
switch(config) # maintenance-unit ospfv2 campus-ospfv2  
switch(config-maintenance-unit-ospfv2-campus-ospfv2) #
```

Creating OSPFv2 maintenance-unit named **campus-ospfv2** with REST API:

(CLI **maintenance-unit ospfv2** maps to **Feature_Set_OSPF** table in REST)

```
curl -X POST "/rest/latest/network/maintenance/feature_set_ospf" \
-H "Content-Type: application/json" \
-d '{"name": "campus-ospfv2"}'
```



NOTE

One **Feature_Set_OSPF** table handles all OSPF variants (v2, v3, v3-AF-IPv4, v3-AF-IPv6)

Delete OSPFv2 maintenance-unit named **campus-ospfv2**

```
switch(config)# no maintenance-unit ospfv2 campus-ospfv2
```

Attempting to delete OSPF maintenance-unit that is applied to a profile will fail:

```
switch(config)# no maintenance-unit ospfv2 campus-ospfv2
```

Error: Cannot delete maintenance-unit 'campus-ospfv2' - currently applied to maintenance-mode profile 'campus-profile'

Remove from profile first, then delete the maintenance-unit:

```
switch(config)# maintenance-mode profile campus-profile
```

```
switch(config-maint-mode-campus-profile)# no maintenance-unit apply campus-ospfv2
```

```
switch(config-maint-mode-campus-profile)# exit
```

```
switch(config)# no maintenance-unit ospfv2 campus-ospfv2
```

REST API Examples

Removing OSPFv2 maintenance-unit:

```
curl -X DELETE "/rest/latest/system/feature_set_ospfs/campus-ospfv2"
```

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config	Administrators or local user group members with execution rights for this command.
6400		
8100		
8325		

Platforms	Command context	Authority
8325H		
8360		
8400		
9300		
9300-32D		
10000		

maintenance-unit ospfv3 <FEATURE_SET_NAME>

Syntax

```
[no] maintenance-unit ospfv3 <FEATURE_SET_NAME>
```

Description

Creates and enters an OSPFv3 feature-set context at the global configuration level. This command allows you to define reusable OSPFv3-specific feature-sets for maintenance mode operations. OSPFv3 maintenance-unit configure max-metric behavior for OSPFv3 processes supporting IPv6 routing. Once created, you can configure OSPFv3 process IDs and VRFs directly (no sub-context). These OSPFv3 maintenance-units can be referenced and applied within maintenance mode profiles.

The **no** form of this command deletes an OSPF maintenance-unit from the global configuration. It also removes the entire OSPF maintenance-unit including all OSPF process and VRF configurations. The OSPF maintenance-unit cannot be deleted if it is currently applied to a maintenance-mode profile.



NOTE

You must first remove the maintenance-unit from any profiles using **no maintenance-unit apply <FEATURE_SET_NAME>** before deletion. The feature-set can be deleted regardless of whether maintenance mode is active, as long as it is not applied to the active profile.

Parameter	Description
<FEATURE_SET_NAME>	Name of the OSPFv3 maintenance-unit. Required

Examples

Creating OSPFv3 maintenance-unit named **campus-ospfv3**:

```
switch(config)# maintenance-unit ospfv3 campus-ospfv3
switch(config-maintenance-unit-ospfv3-campus-ospfv3)#
```

Deleting OSPFv3 feature-set named **campus-ospfv3**

```
switch(config)# no maintenance-unit ospfv3 campus-ospfv3
```

REST API Examples

Creating OSPFv3 maintenance-unit named **campus-ospfv3** with REST API:

(CLI **maintenance-unit ospfv3** maps to **Feature_Set_OSPF** table in REST)

```
curl -X POST "/rest/latest/system/feature_set_ospfs" \
-H "Content-Type: application/json" \
-d '{"name": "campus-ospfv3", "protocol_type": "ospfv3"}'
```

Removing OSPFv3 maintenance-unit

```
curl -X DELETE "/rest/latest/system/feature_set_ospfs/campus-ospfv3"
```

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config	Administrators or local user group members with execution rights for this command.
6400		
8100		
8325		
8325H		
8360		
8400		
9300		
9300-32D		
10000		

maintenance-unit ospfv3-af-ipv4 <FEATURE_SET_NAME>

Syntax

```
[no] maintenance-unit ospfv3-af-ipv4 <FEATURE_SET_NAME>
```

Description

Creates and enters an OSPFv3-AF-IPv4 maintenance-unit context at the global configuration level. This command allows you to define reusable OSPFv3-AF-IPv4-specific maintenance-units for maintenance mode operations. OSPFv3-AF-IPv4 maintenance-units configure max-metric behavior for OSPFv3 processes using the IPv4 address family. Once created, you can configure OSPFv3-AF-IPv4 process IDs and VRFs directly (no sub-context). These OSPFv3-AF-IPv4 maintenance-units can be referenced and applied within maintenance mode profiles.

The **no** form of this command deletes an OSPF maintenance-unit from the global configuration. It also removes the entire OSPF maintenance-unit including all OSPF process and VRF configurations. The OSPF maintenance-unit cannot be deleted if it is currently applied to a maintenance-mode profile.



NOTE

You must first remove the feature-set from any profiles using **no maintenance-unit apply <FEATURE_SET_NAME>** before deletion. The maintenance-unit can be deleted regardless of whether maintenance mode is active, as long as it is not applied to the active profile.

Parameter	Description
<FEATURE_SET_NAME>	Name of the OSPFv3-AF-IPv4 maintenance-unit. Required

Examples

Creating OSPFv3 maintenance-unit named **campus-ospfv3-af-ipv4**:

```
switch(config) # maintenance-unit ospfv3-af-ipv4 campus-ospfv3-af-ipv4  
switch(config-maintenance-unit-ospfv3-af-ipv4-campus-ospfv3-af-ipv4) #
```

Creating OSPFv3-AF-IPv4 maintenance-unit named **campus-ospfv3-af-ipv4** with REST API:

(CLI **maintenance-unit ospfv3-af-ipv4** maps to **Feature_Set_OSPF** table in REST)

```
curl -X POST "/rest/latest/system/feature_set_ospfs" \
-H "Content-Type: application/json" \
-d '{"name": "campus-ospfv3-af-ipv4", "protocol_type": "ospfv3-af-ipv4"}'
```

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config	Administrators or local user group members with execution rights for this command.
6400		
8100		
8325		
8325H		
8360		
8400		
9300		
9300-32D		
10000		

maintenance-unit ospfv3-af-ipv6 <FEATURE_SET_NAME>

Syntax

```
[no] maintenance-unit ospfv3-af-ipv6 <FEATURE_SET_NAME>
```

Description

Creates and enters an OSPFv3-AF-IPv6 maintenance-unit context at the global configuration level. This command allows you to define reusable OSPFv3-AF-IPv6-specific maintenance-units for maintenance mode operations. OSPFv3-AF-IPv6 maintenance-units configure max-metric behavior for OSPFv3 processes using

the IPv6 address family. Once created, you can configure OSPFv3-AF-IPv6 process IDs and VRFs directly (no sub-context). These OSPFv3-AF-IPv6 maintenance-units can be referenced and applied within maintenance mode profiles.

The **no** form of this command deletes an OSPF maintenance-unit from the global configuration. It also removes the entire OSPF maintenance-unit including all OSPF process and VRF configurations. The OSPF maintenance-unit cannot be deleted if it is currently applied to a maintenance-mode profile.



NOTE

You must first remove the maintenance-unit from any profiles using **no maintenance-unit apply <FEATURE_SET_NAME>** before deletion. The maintenance-unit can be deleted regardless of whether maintenance mode is active, as long as it is not applied to the active profile.

Parameter	Description
<FEATURE_SET_NAME>	Name of the OSPFv3-AF-IPv6 maintenance-unit. Required

Examples

Creating OSPFv3-AF-IPv6 feature-set named **campus-ospfv3-af-ipv6**:

```
switch(config)# maintenance-unit ospfv3-af-ipv6 campus-ospfv3-af-ipv6
switch(config-feature-set-ospfv3-af-ipv6-campus-ospfv3-af-ipv6)#
```

Creating OSPFv3-AF-IPv6 maintenance-unit named **campus-ospfv3-af-ipv6** with REST API:

(CLI **maintenance-unit ospfv3-af-ipv6** maps to **Feature_Set_OSPF** table in REST)

```
curl -X POST "/rest/latest/system/feature_set_ospfs" \
-H "Content-Type: application/json" \
-d '{"name": "campus-ospfv3-af-ipv6", "protocol_type": "ospfv3-af-ipv6"}'
```

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config	Administrators or local user group members with execution rights for this command.
6400		
8100		
8325		
8325H		
8360		
8400		
9300		
9300-32D		
10000		

```
neighbor <IP_ADDRESS> vrf <VRF_NAME>
```

Syntax

```
[no] {neighbor <IP_ADDRESS> vrf <VRF_NAME> | all-neighbor}
```

Description

Specifies the BGP neighbor IP address and VRF within a BGP maintenance-unit, or configures all BGP neighbors from all VRFs. You can specify individual neighbor IP addresses with specific VRF names, or use **all-neighbor** to include all BGP neighbors from all VRFs. Multiple individual neighbor commands can be configured within the same BGP maintenance-unit. The **all-neighbor** command overrides any specific neighbor configurations since it includes everything. When applied during maintenance mode, BGP graceful shutdown (GSHUT) community is sent to peers and BGP neighbor sessions are brought down.

The **no** form removes specific BGP configurations from a BGP maintenance-unit. The **no neighbor** command removes a specific BGP neighbor and VRF combination. The **no all-neighbor** command removes the configuration that affects all BGP neighbors across all VRFs. These provide granular control without deleting the entire BGP maintenance-unit.

Parameter	Description
neighbor <IP_ADDRESS>	BGP neighbor IP address (when using specific neighbor).

Parameter	Description
vrf <VRF_NAME>	Specific VRF name (Only with specific neighbor IP).
all-neighbor	All BGP neighbors from all VRFs (alternative to specific neighbor).

Examples

Configuring BGP neighbor 10.1.1.1 in VRF red in the maintenance-unit

```
switch(config) # maintenance-unit bgp campus-bgp
switch(config-maintenance-unit-bgp-campus-bgp) # neighbor 10.1.1.1 vrf red
```

Configuring multiple BGP neighbors in different VRFs in the maintenance-unit:

```
switch(config) # maintenance-unit bgp campus-bgp
switch(config-maintenance-unit-bgp-campus-bgp) # neighbor 10.1.1.1 vrf red
switch(config-maintenance-unit-bgp-campus-bgp) # neighbor 10.1.1.2 vrf red
switch(config-maintenance-unit-bgp-campus-bgp) # neighbor 10.2.1.1 vrf blue
```

Configuring all BGP neighbors from all VRFs in the maintenance-unit:

```
switch(config) # maintenance-unit bgp campus-bgp
switch(config-maintenance-unit-bgp-campus-bgp) # all-neighbor
```

Removing specific BGP neighbor 10.1.1.1 from VRF red

```
switch(config) # maintenance-unit bgp campus-bgp
switch(config-maintenance-unit-bgp-campus-bgp) # no neighbor 10.1.1.1 vrf red
```

Removing all-neighbor configuration

```
switch(config) # maintenance-unit bgp campus-bgp
switch(config-maintenance-unit-bgp-campus-bgp) # no all-neighbor
```

REST API Examples

Configuring specific BGP neighbors with VRFs (using VRF URI references):

```
curl -X PATCH "/rest/latest/system/feature_set_bgps/campus-bgp" \
-H "Content-Type: application/json" \
-d '{
  "bgp_peer_vrfs": {
    "10.1.1.1": "/rest/latest/system/vrfs/red",
    "10.1.1.2": "/rest/latest/system/vrfs/red"
  },
  "shutdown_delay_timer": 120
}'
```

Configuring BGP to affect all neighbors across all VRFs:

```
curl -X PATCH "/rest/latest/system/feature_set_bgps/campus-bgp" \
-H "Content-Type: application/json" \
```

```
-d '{
  "all_instances": true,
  "shutdown_delay_timer": 120
}'
```

Configuring specific peer addresses that apply to all VRFs:

```
curl -X PATCH "/rest/latest/network/maintenance/feature_set_bgp/campus-bgp" \
-H "Content-Type: application/json" \
-d '{
  "bgp_all_vrfs_peer_addresses": ["10.1.1.1", "10.1.1.2"],
  "shutdown_delay_timer": 120
}'
```

Removing specific BGP neighbor/VRF combination. Get current configuration, modify, then update:

```
curl -X GET "/rest/latest/system/feature_set_bgps/campus-bgp" > current.json
```

Edit **current.json** to remove specific neighbor/VRF entry from **bgp_peer_vrfs** map

```
curl -X PATCH "/rest/latest/system/feature_set_bgps/campus-bgp" \
-H "Content-Type: application/json" \
-d '{
  "bgp_peer_vrfs": {
    "10.1.1.2": "/rest/latest/system/vrfs/red"
  }
}'
```

Removing all-neighbor configuration:

```
curl -X PATCH "/rest/latest/system/feature_set_bgps/campus-bgp" \
-H "Content-Type: application/json" \
-d '{"all_instances": false}'
```

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config	Administrators or local user group members with execution rights for this command.
6400		
8100		

Platforms	Command context	Authority
8325		
8325H		
8360		
8400		
9300		
9300-32D		
10000		

process <PROCESS_ID>

Syntax

```
[no] process <PROCESS_ID> {vrf <VRF_NAME> | all-vrfs | all-process }
```

Description

Specifies the OSPF process ID (OSPFv2, OSPFv3, OSPFv3-AF-IPv4 or OSPFv3-AF-IPv6) and VRF within an OSPF maintenance-unit, or configures all OSPF processes from all VRFs. You can specify individual process IDs with specific VRF names, configure a specific process ID across all VRFs using **all-vrfs**, or use **all-process** to include all OSPF processes from all VRFs.



NOTE

OSPFv3-AF-IPv4 and uses OSPFv3 with the IPv4 address family. OSPFv3-AF-IPv6 uses OSPFv3 with the IPv6 address family.

The **no** form of this command removes a specific OSPF process and VRF combination from an OSPF maintenance-unit, removes a process configured with all-vrfs, or removes the all-process configuration. This provides granular control to remove individual OSPF processes without deleting the entire OSPF maintenance-unit. Multiple process commands can be removed individually. The **no process <ID> all-vrfs** command removes the **all-VRFs** configuration for a specific process ID. The **no all-process** command removes the configuration that affects all OSPF processes across all VRFs.

Usage

Configuration Options:

Option	Command	Description
Specific Process + Specific VRF	process vrf <VRF_NAME>	Targets a single process in a single VRF
Specific Process + All VRFs	process <ID> all-vrfs	Applies the specified process ID across all VRFs (including default VRF)
All Processes	process <ID> all-process	Includes all process IDs from all VRFs

Multiple individual process commands can be configured within the same OSPF maintenance-unit. When both specific VRF and **all-vrfs** configurations exist for the same process ID, the **all-vrfs** configuration takes precedence as it is a superset. The **all-process** command includes all process IDs from all VRFs and represents the broadest scope.



NOTE

Important Configuration Timing: Configuration changes made before **maintenance-mode activate** take effect immediately. Configuration changes made after maintenance mode is activated have indeterminate behavior and should be avoided.

When applied during maintenance mode, OSPF processes use the max-metric approach to signal maintenance state to neighboring routers.

Parameter	Description
<PROCESS_ID>	OSPF process ID (when using specific process). Required Range: 1-65535
vrf <VRF_NAME>	Specific VRF name (only with specific process ID and specific VRF).
all-process	All OSPF processes from all VRFs (alternative to specific processes).
all-vrfs	Apply process ID to all VRFs including default VRF (alternative to specific VRF).

Examples

Configuring OSPFv2 process 1 in VRF red:

```
switch(config) # maintenance-unit ospfv2 campus-ospfv2  
switch(config-maintenance-unit-ospfv2-campus-ospfv2) # process 1 vrf red
```

Configuring OSPFv3 process 1 in VRF red:

```
switch(config) # maintenance-unit ospfv3 campus-ospfv3  
switch(config-maintenance-unit-ospfv3-campus-ospfv3) # process 1 vrf red
```

Configuring OSPFv3-AF-IPv4 process 1 in VRF red:

```
switch(config) # maintenance-unitospfv3-af-ipv4 campus-ospfv3-af-ipv4  
switch(config-maintenance-unit-ospfv3-af-ipv4-campus-ospfv3-af-ipv4) #  
process 1 vrf red
```

Configuring OSPFv3-AF-IPv6 process 1 in VRF red:

```
switch(config) # maintenance-unitospfv3-af-ipv6 campus-ospfv3-af-ipv6  
switch(config-maintenance-unit-ospfv3-af-ipv6-campus-ospfv3-af-ipv6) #  
process 1 vrf red
```

For the following examples, replace OSPFv2 with OSPFv3, OSPFv3-af-ipv4 or OSPFv3-af-ipv6 as needed.

Configuring multiple OSPFv2 processes in different VRFs:

```
switch(config) # maintenance-unit ospfv2 campus-ospfv2  
switch(config-maintenance-unit-ospfv2-campus-ospfv2) # process 1 vrf red  
switch(config-maintenance-unit-ospfv2-campus-ospfv2) # process 2 vrf red  
switch(config-maintenance-unit-ospfv2-campus-ospfv2) # process 3 vrf blue
```

Configuring OSPFv2 process 1 across all VRFs:

```
switch(config) # maintenance-unit ospfv2 campus-ospfv2  
switch(config-maintenance-unit-ospfv2-campus-ospfv2) # process 1 all-vrfs
```

Configuring multiple OSPFv2 processes across all VRFs:

```
switch(config) # maintenance-unit ospfv2 campus-ospfv2  
switch(config-maintenance-unit-ospfv2-campus-ospfv2) # process 1 all-vrfs  
switch(config-maintenance-unit-ospfv2-campus-ospfv2) # process 2 all-vrfs
```

Configuring all OSPFv2 processes from all VRFs

```
switch(config) # maintenance-unitospfv2 campus-ospfv2  
switch(config-maintenance-unit-ospfv2-campus-ospfv2) # all-process
```

Removing OSPFv2 process 1 from VRF red:

```
switch(config) # maintenance-unitospfv2 campus-ospfv2  
switch(config-maintenance-unit-ospfv2-campus-ospfv2) # no process 1 vrf red
```

Removing OSPFv2 process 1 **all-vrfs** configuration:

```
switch(config) # maintenance-unit ospfv2 campus-ospfv2  
switch(config-maintenance-unit-ospfv2-campus-ospfv2) # no process 1 all-vrfs
```

Removing all-process configuration

```
switch(config) # maintenance-unit ospfv2 campus-ospfv2  
switch(config-maintenance-unit-ospfv2-campus-ospfv2) # no all-process
```

REST API Examples



NOTE

The OSPFv2 examples also apply to OSPFv3. Replace **campus-ospfv2** with **campus-ospfv3**.

Configuring OSPFv2 processes with specific VRFs (using VRF URI references):

```
curl -X PATCH "/rest/latest/system/feature_set_ospfs/campus-ospfv2" \
-H "Content-Type: application/json" \
-d '{
  "ospf_process_vrfs": {
    "1": "/rest/latest/system/vrfs/red",
    "2": "/rest/latest/system/vrfs/red"
  }
}'
```

Configuring OSPFv2 process 1 to apply across all VRFs (including default):

```
curl -X PATCH "/rest/latest/system/feature_set_ospfs/campus-ospfv2" \
-H "Content-Type: application/json" \
-d '{
  "ospf_all_vrfs_process_ids": [1]
}'
```

Configuring multiple OSPFv2 processes to apply across all VRFs:

```
curl -X PATCH "/rest/latest/system/feature_set_ospfs/campus-ospfv2" \
-H "Content-Type: application/json" \
-d '{
  "ospf_all_vrfs_process_ids": [1, 2]
}'
```

Configuring OSPFv2 to affect all processes across all VRFs:

```
curl -X PATCH "/rest/latest/system/feature_set_ospfs/campus-ospfv2" \
-H "Content-Type: application/json" \
-d '{"all_instances": true}'
```

Configuring specific OSPFv2 process IDs that apply to all VRFs:

```
curl -X PATCH "/rest/latest/system/feature_set_ospfs/campus-ospfv2" \
-H "Content-Type: application/json" \
-d '{
  "ospf_all_vrfs_process_ids": [1, 2, 3]
}'
```

Configuring specific OSPFv3-AF-IPv4 processes with specific VRF:

```
curl -X PATCH "/rest/latest/system/feature_set_ospfs/campus-ospfv3-af-ipv4" \
-H "Content-Type: application/json" \
```

```
-d '{
  "ospf_process_vrfs": {
    "1": "/rest/latest/system/vrfs/red"
  }
}'
```

Configuring OSPFv3-AF-IPv4 process 1 to apply across all VRFs (including default)

```
curl -X PATCH "/rest/latest/system/feature_set_ospfs/campus-ospfv3-af-ipv4" \
-H "Content-Type: application/json" \
-d '{
  "ospf_all_vrfs_process_ids": [1]
}'
```

Configuring specific OSPFv3-AF-IPv6 process/VRF combination:

```
curl -X PATCH "/rest/latest/system/feature_set_ospfs/campus-ospfv3-af-ipv6" \
-H "Content-Type: application/json" \
-d '{
  "ospf_process_vrfs": {
    "1": "/rest/latest/system/vrfs/red"
  }
}'
```

Configuring multiple OSPFv3-AF-IPv6 processes:

```
curl -X PATCH "/rest/latest/system/feature_set_ospfs/campus-ospfv3-af-ipv6" \
-H "Content-Type: application/json" \
-d '{
  "ospf_process_vrfs": {
    "1": "/rest/latest/system/vrfs/red",
    "2": "/rest/latest/system/vrfs/red"
  }
}'
```

Configuring OSPFv3-AF-IPv6 process 1 to apply across all VRFs (including default)

```
curl -X PATCH "/rest/latest/system/feature_set_ospfs/campus-ospfv3-af-ipv6" \
-H "Content-Type: application/json" \
-d '{
  "ospf_all_vrfs_process_ids": [1]
}'
```

Configuring OSPFv3-AF-IPv6 to affect all processes across all VRFs:

```
curl -X PATCH "/rest/latest/system/feature_set_ospfs/campus-ospfv3-af-ipv6" \
```

```
-H "Content-Type: application/json" \
-d '{"all_instances": true}'
```

To remove specific OSPF process/VRF combination using REST API, get current configuration, modify, then update:

```
curl -X GET "/rest/latest/system/feature_set_ospfs/campus-ospfv2" >
current.json
```

Edit **current.json** to remove specific process/VRF entry from **ospf_process_vrfs** map, then update:

```
curl -X PATCH "/rest/latest/system/feature_set_ospfs/campus-ospfv2" \
-H "Content-Type: application/json" \
-d '{
  "ospf_process_vrfs": {
    "2": "/rest/latest/system/vrfs/red"
  }
}'
```

Removing process all-vrfs configuration:

```
curl -X PATCH "/rest/latest/system/feature_set_ospfs/campus-ospfv2" \
-H "Content-Type: application/json" \
-d '{
  "ospf_process_vrfs": {
    "1": []
  }
}'
```

Removing all-process configuration:

```
curl -X PATCH "/rest/latest/system/feature_set_ospfs/campus-ospfv2" \
-H "Content-Type: application/json" \
-d '{"all_instances": false}'
```

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config	Administrators or local user group members with execution rights for this command.
6400		
8100		

Platforms	Command context	Authority
-----------	-----------------	-----------

8325

8325H

8360

8400

9300

9300-32D

10000

show events -d gshut-daemon

Syntax

```
show events -d gshut-daemon
```

Description

Displays event logs generated by **gshut-daemon** for maintenance mode operations. Use this command to monitor the progress of maintenance mode activation and deactivation, and to diagnose graceful shutdown sequencing for BGP and OSPF protocols.

Example

Displaying event logs during a BGP and OSPF maintenance mode activation:

```
switch# show events -d gshut-daemon

-----
show event logs
-----
2026-03-08:10:00:01.000000|gshut-daemon|17601|LOG_INFO|UKWN|-|Maintenance
mode activated using profile 'campus-maintenance-profile'
2026-03-08:10:00:01.050000|gshut-daemon|17604|LOG_INFO|UKWN|-|BGP 'bgp-
campus-fs': graceful shutdown community advertised to all neighbors
2026-03-08:10:00:01.100000|gshut-daemon|17605|LOG_INFO|UKWN|-|BGP graceful
shutdown timer started for 'bgp-campus-fs' (120 seconds)
2026-03-08:10:02:01.150000|gshut-daemon|17606|LOG_INFO|UKWN|-|BGP graceful
shutdown timer for 'bgp-campus-fs' expired (timer elapsed), triggering
neighbor session shutdown
2026-03-08:10:02:01.200000|gshut-daemon|17604|LOG_INFO|UKWN|-|BGP 'bgp-
```

```
campus-fs': all neighbors gracefully shutdown
2026-03-08:10:02:01.300000|gshut-daemon|17603|LOG_INFO|UKWN|-|BGP process
'bgp-campus-fs' graceful shutdown executed, neighbors in maintenance state
2026-03-08:10:02:01.400000|gshut-daemon|17603|LOG_INFO|UKWN|-|OSPF process
'ospf-campus-fs' graceful shutdown executed, neighbors in maintenance state
```

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config	Administrators or local user group members with execution rights for this command.
6400		
8100		
8325		
8325H		
8360		
8400		
9300		
9300-32D		
10000		

show maintenance-mode profile

Syntax

```
show maintenance-mode profile
```

Description

Displays the configuration details of the maintenance mode profile including maintenance-unit breakdown and protocol-specific settings. This command shows what is configured, not runtime state. Use this command to review the maintenance mode configuration including BGP shutdown timers and OSPF process selections. The output clearly distinguishes between default (built-in) and custom profiles.

Parameter	Description
profile	Command to show profile configuration. Required

Examples

Displaying maintenance mode profile when default profile is used:

```
switch# show maintenance-mode profile
Maintenance Mode Profile: default (built-in)

Applied Feature-Sets:
- default-feature

BGP Configuration:
  Shutdown Delay Timer: 120 seconds
  Neighbor Scope: All neighbors from all VRFs

OSPF Configuration:
  OSPFv2: All processes from all VRFs
  OSPFv3: All processes from all VRFs
  OSPFv3-AF-IPv4: All processes from all VRFs
  OSPFv3-AF-IPv6: All processes from all VRFs
```



NOTE

Default profile does not appear in running-config

Displaying custom maintenance mode profile configuration:

```
switch# show maintenance-mode profile
=== ACTIVE PROFILE (Custom) ===
Maintenance Mode Profile: campus-profile

Applied Feature-Sets:
- campus-feature

BGP Configuration:
  Shutdown Delay Timer: 120 seconds
  Neighbors:
    - 10.1.1.1 (VRF: red)
    - 10.1.1.2 (VRF: red)
```



```
OSPF Configuration:
  OSPFv2 Processes:
    - Process 1 (VRF: red)
    - Process 2 (VRF: red)
  OSPFv3 Processes:
    - Process 1 (VRF: red)
```

=== DEFAULT PROFILE (Reference) ===

Note: This built-in profile would be used if custom profile is deleted

```
Applied Feature-Sets:
- default-feature
```

```
BGP Configuration:
  Shutdown Delay Timer: 120 seconds
  Neighbor Scope: All neighbors from all VRFs
```

```
OSPF Configuration:
  OSPFv2: All processes from all VRFs
  OSPFv3: All processes from all VRFs
  OSPFv3-AF-IPv4: All processes from all VRFs
  OSPFv3-AF-IPv6: All processes from all VRFs
```

Displaying profile with multiple maintenance-units:

```
switch# show maintenance-mode profile
=== ACTIVE PROFILE (Custom) ===
Maintenance Mode Profile: complete-profile
```

```
Applied Feature-Sets:
- underlay-feature
```

```
OSPF Configuration:
  OSPFv2 Processes:
    - Process 1 (VRF: red)
    - Process 2 (VRF: red)
  OSPFv3 Processes:
    - Process 1 (VRF: red)
```

```
- overlay-feature
```

```
BGP Configuration:
  Shutdown Delay Timer: 120 seconds
  Neighbors:
    - 10.1.1.1 (VRF: red)
```

```
- 10.1.1.2 (VRF: red)
```

```
=== DEFAULT PROFILE (Reference) ===
```

Note: This built-in profile would be used if custom profile is deleted

Applied Feature-Sets:

```
- default-feature
```

BGP Configuration:

Shutdown Delay Timer: 120 seconds

Neighbor Scope: All neighbors from all VRFs

OSPF Configuration:

OSPFv2: All processes from all VRFs

OSPFv3: All processes from all VRFs

OSPFv3-AF-IPv4: All processes from all VRFs

OSPFv3-AF-IPv6: All processes from all VRFs

Displaying profile when no profile is configured

```
switch# show maintenance-mode profile
```

No custom maintenance mode profile configured



NOTE

By default, when maintenance mode is activated without a profile, the built-in default profile will be applied, affecting all OSPF processes and all BGP neighbors.

To view default profile details run **maintenance-mode activate** without profile parameter to apply default, then run **show maintenance-mode profile** to see active default configuration.

REST API Examples

Step 1: Get mode profile with all details:

```
curl -X GET "/rest/latest/system/maintenance_mode_profiles/campus-mode-profile"
```

Step 2: Get all BGP maintenance-units:

```
curl -X GET "/rest/latest/system/feature_set_bgps"
```

Step 3: Get all OSPF maintenance-units

```
curl -X GET "/rest/latest/system/feature_set_ospfs"
```

Step 4: Get specific maintenance-unit configuration:

```
curl -X GET "/rest/latest/system/feature_set_bgps/campus-bgp"
```

```
curl -X GET "/rest/latest/system/feature_set_ospfs/campus-ospfv2"
```

Step 5: Get stage configuration with linked maintenance-units:

```
curl -X GET "/rest/latest/system/maintenance_mode_profiles/campus-mode-profile/stages/1"
```



NOTE

The **maintenance-unit {bgp | ospfv2 | ospfv3}** command maps to two separate REST tables:

Command	REST table
Feature_Set_BGP	/rest/latest/system/feature_set_bgps
Feature_Set_OSPF	'/rest/latest/system/feature_set_ospfs' (handles all OSPF variants)

The **maintenance-mode profile** command maps to **Maintenance_Mode_Profile** in REST API.

Feature-sets are referenced by URI paths in separate **feature_set_bgp** and **feature_set_ospf** arrays on the stage

VRF references use URI paths. For example **/rest/latest/system/vrfs/red**.

Stages must be explicitly created via POST after creating the mode profile

Each protocol has its own dedicated table with protocol-specific fields.

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config	Administrators or local user group members with execution rights for this command.
6400		
8100		
8325		
8325H		
8360		

8400

9300

9300-32D

10000

show maintenance-mode

Syntax

```
show maintenance-mode
```

Description

Displays comprehensive maintenance mode operational status including current state and protocol execution details. This command provides real-time visibility into maintenance mode operations including protocol-specific status information. Use this command to monitor the operational state of an active maintenance session.

Examples

Displaying maintenance mode status when active with custom profile:

```
switch# show maintenance-mode
Maintenance Mode Status: Active
Active Profile: campus-profile
Activation Time: 14:32:15
Duration: 00:15:42

Protocol Status:
  BGP:
    Status: GSHUT community advertised
    Shutdown Timer: 42 seconds remaining (of 120 configured)
    Neighbors Affected: 2 (VRF red)

  OSPF:
    OSPFv2: max-metric active (2 processes in VRF red)
    OSPFv3: max-metric active (1 process in VRF red)

Applied Feature-Sets:
  - campus-feature (Active)
```

Displaying maintenance mode status when active with default profile:

```
switch# show maintenance-mode
Maintenance Mode Status: Active
Active Profile: default
Activation Time: 14:32:15
Duration: 00:15:42

Protocol Status:
  BGP:
    Status: GSHUT community advertised
    Shutdown Timer: 42 seconds remaining (of 120 configured)
    Neighbors Affected: All neighbors from all VRFs

  OSPF:
    OSPFv2: max-metric active (all processes from all VRFs)
    OSPFv3: max-metric active (all processes from all VRFs)

Applied Feature-Sets:
  - default-feature (Active)
```

Displaying maintenance mode status during deactivation:

```
switch# show maintenance-mode
Maintenance Mode Status: Deactivating
Active Profile: campus-profile
Deactivation Started: 15:45:10
Deactivation Duration: 00:02:15

Protocol Restoration Status:
  BGP: Restoring neighbor sessions (2 neighbors)
  OSPF: Removing max-metric (3 processes)

Applied Feature-Sets:
  - campus-feature (Restoring)
```

Displaying maintenance mode status when not active:

```
switch# show maintenance-mode
Maintenance Mode Status: Inactive
Active Profile: None
```

REST API Examples

Step 1: Get mode profile status (includes activated state, maintenance_status):

```
curl -X GET "/rest/latest/system/maintenance_mode_profiles/campus-mode-profile"
```

Step 2: Get all BGP maintenance-units:

```
curl -X GET "/rest/latest/system/feature_set_bgps"
```

Step 3: Get all OSPF maintenance-units:

```
curl -X GET "/rest/latest/system/feature_set_ospfs"
```

Step 4: Get specific BGP maintenance-unit details:

```
curl -X GET "/rest/latest/system/feature_set_bgps/campus-bgp"
```

Step 5: Get stage details with maintenance-unit references:

```
curl -X GET "/rest/latest/system/maintenance_mode_profiles/campus-mode-profile/stages/1"
```

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config	Administrators or local user group members with execution rights for this command.
6400		
8100		
8325		
8325H		
8360		
8400		
9300		
9300-32D		
10000		

show maintenance-unit

Syntax

```
show maintenance-unit {bgp | ospf}
```

Description

Displays all configured maintenance-units with optional filtering by protocol type. This command shows both default (built-in) and user-configured maintenance-units along with their key configuration parameters. Use this command to review all available maintenance-units that can be applied to maintenance-mode profiles. When run without a protocol filter, both BGP and OSPF maintenance-units are displayed. Use the **bgp** keyword to display only BGP maintenance-units. Use the **ospf** keyword to display only OSPF maintenance-units (including OSPFv2, OSPFv3, OSPFv3-AF-IPv4, and OSPFv3-AF-IPv6).

Parameter	Description
{bgp ospf}	Name of the protocol to filter the output. Optional

Examples

Displaying all maintenance-units when no custom maintenance-units are configured:

```
switch# show maintenance-unit
Maintenance Units:
=====

Default Maintenance-Units:
- default-bgp (Default, BGP)
  Shutdown Delay Timer: 120 seconds
  Neighbor Scope: All neighbors from all VRFs
- default-ospfv2 (Default, OSPFv2)
  Process Scope: All processes from all VRFs
- default-ospfv3-af-ipv6 (Default, OSPFv3-AF-IPv6)
  Process Scope: All processes from all VRFs
- default-ospfv3-af-ipv4 (Default, OSPFv3-AF-IPv4)
  Process Scope: All processes from all VRFs
- default-ospfv3 (Default, OSPFv3)
  Process Scope: All processes from all VRFs
```

Displaying only BGP maintenance-units when no custom BGP maintenance-units are configured

```
switch# show maintenance-unit bgp
Maintenance Units:
=====
```

Default Maintenance-Units:

- default-bgp (Default, BGP)
Shutdown Delay Timer: 120 seconds
Neighbor Scope: All neighbors from all VRFs

User-configured Maintenance-Units:

No custom BGP maintenance-unit configured.

Displaying only OSPF maintenance-units when no custom OSPF maintenance-units are configured:

```
switch# show maintenance-unit ospf
```

Maintenance Units:

=====

Default Maintenance-Units:

- default-ospfv2 (Default, OSPFv2)
Process Scope: All processes from all VRFs
- default-ospfv3-af-ipv6 (Default, OSPFv3-AF-IPv6)
Process Scope: All processes from all VRFs
- default-ospfv3-af-ipv4 (Default, OSPFv3-AF-IPv4)
Process Scope: All processes from all VRFs
- default-ospfv3 (Default, OSPFv3)
Process Scope: All processes from all VRFs

User-configured Maintenance-Units:

No custom OSPF maintenance-unit configured.

Displaying all maintenance-units with custom maintenance-units configured:

```
switch# show maintenance-unit
```

Maintenance Units:

=====

Default Maintenance-Units:

- default-bgp (Default, BGP)
Shutdown Delay Timer: 120 seconds
Neighbor Scope: All neighbors from all VRFs
- default-ospfv2 (Default, OSPFv2)
Process Scope: All processes from all VRFs
- default-ospfv3-af-ipv6 (Default, OSPFv3-AF-IPv6)
Process Scope: All processes from all VRFs
- default-ospfv3-af-ipv4 (Default, OSPFv3-AF-IPv4)
Process Scope: All processes from all VRFs
- default-ospfv3 (Default, OSPFv3)
Process Scope: All processes from all VRFs

User-configured Maintenance-Units:

- campus-bgp (Custom, BGP)
Shutdown Delay Timer: 180 seconds
Neighbors:
 - 10.1.1.1 (VRF: red)
 - 10.1.1.2 (VRF: red)
- campus-ospfv2 (Custom, OSPFv2)
Processes:
 - Process 1 (VRF: red)
 - Process 2 (VRF: red)

Displaying only BGP maintenance-units with custom BGP maintenance-unit configured:

```
switch# show maintenance-unit bgp
```

Maintenance Units:

=====

Default Maintenance-Units:

- default-bgp (Default, BGP)
Shutdown Delay Timer: 120 seconds
Neighbor Scope: All neighbors from all VRFs

User-configured Maintenance-Units:

- campus-bgp (Custom, BGP)
Shutdown Delay Timer: 180 seconds
Neighbors:
 - 10.1.1.1 (VRF: red)
 - 10.1.1.2 (VRF: red)

Displaying only OSPF maintenance-units with custom OSPF maintenance-unit configured:

```
switch# show maintenance-unit ospf
```

Maintenance Units:

=====

Default Maintenance-Units:

- default-ospfv2 (Default, OSPFv2)
Process Scope: All processes from all VRFs
- default-ospfv3-af-ipv6 (Default, OSPFv3-AF-IPv6)
Process Scope: All processes from all VRFs
- default-ospfv3-af-ipv4 (Default, OSPFv3-AF-IPv4)
Process Scope: All processes from all VRFs
- default-ospfv3 (Default, OSPFv3)
Process Scope: All processes from all VRFs

User-configured Maintenance-Units:

- campus-ospfv2 (Custom, OSPFv2)

```
Processes:
- Process 1 (VRF: red)
- Process 2 (VRF: red)
```

Display when no maintenance-units are configured:

```
switch# show maintenance-unit
No maintenance-units configured
```

Display when only user-configured maintenance-units exist for a protocol:

```
switch# show maintenance-unit bgp
Maintenance Units:
=====

User-configured Maintenance-Units:
- campus-bgp (User-configured, BGP)
  Shutdown Delay Timer: 120 seconds (default)
  Neighbor Scope: All neighbors from all VRFs
```

REST API Example

Get all BGP maintenance-units (equivalent to **show maintenance-unit bgp**):

```
curl -X GET "/rest/latest/network/maintenance/feature_set_bgp"
```

Get all OSPF maintenance-units (equivalent to **show maintenance-unit ospf**):

```
curl -X GET "/rest/latest/network/maintenance/feature_set_ospf"
```

Get specific BGP maintenance-unit details:

```
curl -X GET "/rest/latest/network/maintenance/feature_set_bgp/campus-bgp"
```

Get specific OSPF maintenance-unit details:

```
curl -X GET "/rest/latest/network/maintenance/feature_set_ospf/campus-ospfv2"
```



NOTE

The response includes origin field to distinguish built-in from configured profiles:

Table 1.

Origin field	maintenance-unit
built-in	Default maintenance-unit
configuration	User-configured maintenance-unit

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config	Administrators or local user group members with execution rights for this command.
6400		
8100		
8325		
8325H		
8360		
8400		
9300		
9300-32D		
10000		

shutdown-delay-timer <SECONDS>

Syntax

```
[no] shutdown-delay-timer <SECONDS>
```

Description

Configures the BGP shutdown delay timer within a BGP maintenance-unit. This timer controls the delay between GSHUT community advertisement and actual BGP neighbor session shutdown during maintenance mode activation. The timer allows BGP peers sufficient time to reconverge and select alternate paths before sessions are brought down. the default value is 120 seconds.

Timer Behavior: When maintenance mode is activated, BGP first advertises the GSHUT community (65535:0) with all routes. After the configured shutdown-delay-timer expires, BGP neighbor sessions are gracefully brought down. This two-phase approach minimizes traffic disruption by allowing peers to preemptively select alternate paths before session teardown.

Timer Range: The shutdown-delay-timer can be configured from 0 to 3600 seconds (1 hour). A value of 0 causes immediate session shutdown without delay after GSHUT community advertisement. Higher values provide more time for network convergence but extend the maintenance mode activation duration.

Scope: The shutdown-delay-timer applies to all BGP neighbors configured within the same BGP maintenance-unit. Different maintenance-units can have different shutdown-delay-timer values for different maintenance scenarios.

The **no** form resets the timer to the default value of 120 seconds. This provides granular control without deleting the entire BGP maintenance-unit.

Parameter	Description
<SECONDS>	Delay in seconds. Range: 0-3600 Default: 120 Required

Examples

Configuring BGP shutdown-delay-timer to 180 seconds with specific neighbors:

```
switch(config) # maintenance-unit bgp campus-bgp
switch(config-maintenance-unit-bgp-campus-bgp) # shutdown-delay-timer 180
switch(config-maintenance-unit-bgp-campus-bgp) # neighbor 10.1.1.1 vrf red
```

Configuring BGP shutdown-delay-timer to default 120 seconds with all neighbors:

```
switch(config) # maintenance-unitbgp campus-bgp
switch(config-maintenance-unit-bgp-campus-bgp) # shutdown-delay-timer 120
switch(config-maintenance-unit-bgp-campus-bgp) # all-neighbor
```

Configuring BGP with custom timer and neighbors:

```
switch(config) # maintenance-unitbgp campus-bgp
switch(config-maintenance-unit-bgp-campus-bgp) # shutdown-delay-timer 300
switch(config-maintenance-unit-bgp-campus-bgp) # neighbor 10.1.1.1 vrf red
switch(config-maintenance-unit-bgp-campus-bgp) # neighbor 10.1.1.2 vrf red
```

Configuring immediate BGP shutdown (0 second delay)

```
switch(config) # maintenance-unit bgp campus-bgp
switch(config-maintenance-unit-bgp-campus-bgp) # shutdown-delay-timer 0
switch(config-maintenance-unit-bgp-campus-bgp) # all-neighbor
```

Resetting BGP shutdown-delay-timer to default:

```
switch(config) # maintenance-unit bgp campus-bgp
switch(config-maintenance-unit-bgp-campus-bgp) # no shutdown-delay-timer
```

REST API Examples

Configuring BGP shutdown-delay-timer to 180 seconds with specific neighbors:

```
curl -X PATCH "/rest/latest/system/feature_set_bgps/campus-bgp" \
-H "Content-Type: application/json" \
-d '{"shutdown_delay_timer": 180}'
```

Configuring BGP shutdown-delay-timer to default 120 seconds with all neighbors:

```
curl -X PATCH "/rest/latest/system/feature_set_bgps/campus-bgp" \
-H "Content-Type: application/json" \
-d '{"shutdown_delay_timer": 120}'
```

Configuring BGP with custom timer:

```
curl -X PATCH "/rest/latest/system/feature_set_bgps/campus-bgp" \
-H "Content-Type: application/json" \
-d '{"shutdown_delay_timer": 300}'
```

Resetting BGP shutdown-delay-timer to default:

```
curl -X PATCH "/rest/latest/system/feature_set_bgps/campus-bgp" \
-H "Content-Type: application/json" \
-d '{"shutdown_delay_timer": 120}'
```

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
6300	config	Administrators or local user group members with execution rights for this command.
6400	config-feature-set-bgp-c	
8100	ampus-bgp	
8325		
8325H		
8360		
8400		
9300		
9300-32D		

Support and other resources

Subtopics

Accessing Hewlett Packard Enterprise Support

HPE product registration

Accessing updates

Remote support

Warranty information

Regulatory information

Documentation feedback

Accessing Hewlett Packard Enterprise Support

- For live assistance, go to the Contact Hewlett Packard Enterprise Worldwide website:

<https://www.hpe.com/info/assistance>

- To access documentation and support services, go to the Hewlett Packard Enterprise Support Center website:

<https://www.hpe.com/support/hpesc>

Information to collect

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components

- Third-party products or components

HPE product registration

To gain the full benefits of the Hewlett Packard Enterprise Support Center and your purchased support services, add your contracts and products to your account on the HPESC.

- When you add your contracts and products, you receive enhanced personalization, workspace alerts, insights through the dashboards, and easier management of your environment.
- You will also receive recommendations and tailored product knowledge to self-solve any issues, as well as streamlined case creation for faster time to resolution when you must create a case.

To learn how to add your contracts and products, see <https://www.hpe.com/info/add-products-contracts>.

Accessing updates

- Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.
- To download product updates:

Hewlett Packard Enterprise Support Center

<https://www.hpe.com/support/hpesc>

My HPE Software Center

<https://www.hpe.com/software/hpesoftwarecenter>

- To subscribe to eNewsletters and alerts:

<https://www.hpe.com/support/e-updates>

- To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:



IMPORTANT

Access to some updates might require product entitlement when accessed through the Hewlett Packard Enterprise Support Center. You must have an HPE Account set up with relevant entitlements.

Remote support

Remote support is available with supported devices as part of your warranty or contractual support agreement. It provides intelligent event diagnosis, and automatic, secure submission of hardware event notifications to Hewlett Packard Enterprise, which initiates a fast and accurate resolution based on the service level of your product. Hewlett Packard Enterprise strongly recommends that you register your device for remote support.

If your product includes additional remote support details, use search to locate that information.

HPE Get Connected

<https://www.hpe.com/services/getconnected>

HPE Tech Care Service

<https://www.hpe.com/services/techcare>

HPE Complete Care Service

<https://www.hpe.com/services/completecure>

Warranty information

To view the warranty information for your product, see the [warranty check tool](#).

Regulatory information

To view the regulatory information for your product, view the Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products, available at the Hewlett Packard Enterprise Support Center:

<https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

Additional regulatory information

Hewlett Packard Enterprise is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements such as REACH (Regulation EC No 1907/2006 of the European Parliament and the Council). A chemical information report for this product can be found at:

<https://www.hpe.com/info/reach>

For Hewlett Packard Enterprise product environmental and safety information and compliance data, including RoHS and REACH, see:

<https://www.hpe.com/info/ecodata>

For Hewlett Packard Enterprise environmental information, including company programs, product recycling, and energy efficiency, see:

<https://www.hpe.com/info/environment>

Documentation feedback

Hewlett Packard Enterprise is committed to providing documentation that meets your needs. To help us improve the documentation, click the **Feedback** button on the page of an opened document on the Hewlett Packard Enterprise Support Center portal (**<https://www.hpe.com/support/hpesc>**). Use this feature to send any errors, suggestions, or comments. This process captures all document information.